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Investor Presentation | Asia Pacific

# Greater China Technology Hardware: Building Blocks of Compute: Servers, AI Infrastructure, and Electronic Components

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GREATER CHINA TECHNOLOGY HARDWARE

Asia Pacific

Industry View

In-Line

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## Investment Summary

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- **Opportunities in AI GPU and ASIC server/rack design upgrades**

- Major design upgrades for upcoming Vera Rubin platform, Kyber architecture and LPX/Vera CPU rack.
- AI ASIC server upgrade and volume expansion magnitude – mainly TPU and Trainium platform.

- **Where we see room for share price upside, near term**

- AI server power solution (800 VDC), PCB/ABF substrate supply tightness, data network/interconnect/CPO upgrades.
- Smooth GB300 server rack delivery plus attractive risk-reward for AI server ODMs.
- Capacity expansion wave across the tech hardware supply chain.

- **What are the potential risks that we observe**

- Consumer electronics (smartphones/PC) demand and margin in 2H impacted by memory inflation.
- Raw material price hikes (copper, nickel) and supply tightness a margin headwind.
- Supply shortage impact could potentially delay the shipment pace.
- Geopolitical tension that impacts AI data center infrastructure spending.

- **Key stock ideas**

- AI server hardware – Wistron, FII/Hon Hai, Wiwynn, Quanta, Delta Electronics, AVC, BizLink, King Slide, Accton, Chenbro, Gold Circuit, Unimicron, NYPCB, Zhen Ding, Yageo, FIT, and Fositek.
- Edge AI – Xiaomi, Luxshare, Lenovo.

## Hardware Technology Valuation Comparison

| Ticker                                 | Company         | Closing Price | Rating | Price Target | Market Cap | EPS (Local Dollar) |          | P/E (X) |      | P/B (X) |        | P/S (X) |      | EV/EBITDA |       | ROE (%) |       | ROA (%)   |       | Trading Volume |     |
|----------------------------------------|-----------------|---------------|--------|--------------|------------|--------------------|----------|---------|------|---------|--------|---------|------|-----------|-------|---------|-------|-----------|-------|----------------|-----|
|                                        |                 | 7/13/2026     |        | (LC)         | (US\$m)    | 26E                | 27E      | 26E     | 27E  | 26E     | 27E    | 26E     | 27E  | 26E       | 27E   | 26E     | 27E   | (US\$ mn) | 26E   |                |     |
| PC / Data Center / Cyclical Components |                 |               |        |              |            |                    |          |         |      |         |        |         |      |           |       |         |       |           |       |                |     |
| Taiwan                                 |                 |               |        |              |            |                    |          |         |      |         |        |         |      |           |       |         |       |           |       |                |     |
| 2324.TW                                | Compal          | 36.65         | U      | 23.00        | 5,020      | 1.98 e             | 2.23 e   | 18.5    | 16.5 | 1.1 e   | 1.0 e  | 0.2     | 0.2  | 8.1       | 7.7   | 5.7     | 6.2   | 1.9       | 2.0   | 51.3           | 2%  |
| 2327.TW                                | Yageo           | 816.00        | O      | 1515.0       | 52,538     | 20.59 e            | 35.13 e  | 39.6    | 23.2 | 7.7 e   | 6.3 e  | 9.0     | 6.5  | 26.6      | 16.0  | 21.5    | 29.8  | 10.0      | 14.6  | 466.5          | 1%  |
| 2353.TW                                | Acer            | 31.95         | U      | 21.00        | 3,027      | 1.54 e             | 1.61 e   | 20.8    | 19.8 | 1.2 e   | 1.1 e  | 0.3     | 0.3  | 9.1       | 8.4   | 5.3     | 6.0   | 1.9       | 2.0   | 28.5           | 4%  |
| 2357.TW                                | Asustek         | 704.00        | U      | 570.0        | 16,253     | 59.22 e            | 57.19 e  | 11.9    | 12.3 | 1.7 e   | 1.7 e  | 0.6     | 0.5  | 8.7       | 9.3   | 14.9    | 13.8  | 7.0       | 6.1   | 87.0           | 6%  |
| 2368.TW                                | GCE             | 1,045.00      | O      | 1660.00      | 16,794     | 47.19 e            | 82.32 e  | 22.1    | 12.7 | 10.0 e  | 6.4 e  | 5.0     | 3.3  | 14.9      | 8.3   | 54.7    | 61.5  | 24.2      | 28.4  | 225.6          | 1%  |
| 2376.TW                                | Gigabyte        | 335.50        | E      | 375.0        | 6,986      | 32.40 e            | 29.75 e  | 10.4    | 11.3 | 3.1 e   | 3.0 e  | 0.4     | 0.4  | 7.3       | 8.2   | 32.4    | 27.1  | 12.3      | 9.7   | 61.9           | 4%  |
| 2382.TW                                | Quanta          | 378.00        | O      | 385.00       | 45,382     | 24.66 e            | 28.80 e  | 15.3    | 13.1 | 4.7 e   | 3.9 e  | 0.4     | 0.4  | 12.4      | 10.4  | 33.6    | 32.3  | 5.6       | 5.1   | 214.4          | 3%  |
| 3231.TW                                | Wistron         | 143.50        | O      | 210.0        | 14,185     | 13.95 e            | 18.11 e  | 10.3    | 7.9  | 1.5 e   | 1.4 e  | 0.1     | 0.1  | 3.1       | 2.5   | 15.8    | 18.2  | 3.5       | 3.7   | 200.6          | 3%  |
| 3533.TW                                | Lotes           | 1,980.00      | E      | 2350.00      | 6,926      | 94.03 e            | 116.81 e | 21.1    | 17.0 | 4.7 e   | 4.0 e  | 5.0     | 4.3  | 12.2      | 9.7   | 24.1    | 25.7  | 19.2      | 20.3  | 68.7           | 2%  |
| 4938.TW                                | Pegatron        | 82.60         | E      | 85.0         | 6,886      | 5.39 e             | 7.61 e   | 15.3    | 10.8 | 0.8 e   | 0.8 e  | 0.2     | 0.1  | 4.1       | 3.2   | 5.6     | 7.5   | 2.2       | 2.6   | 27.8           | 0%  |
| 4958.TW                                | Zhen Ding       | 646.00        | O      | 690.00       | 22,265     | 15.08 e            | 25.30 e  | 42.8    | 25.5 | 3.8 e   | 3.4 e  | 3.1     | 2.4  | 18.6      | 12.4  | 9.1     | 14.0  | 5.5       | 8.5   | 205.0          | 1%  |
| 8046.TW                                | Nan Ya PCB      | 1,270.00      | O      | 1550.0       | 25,507     | 15.64 e            | 33.02 e  | 81.2    | 38.5 | 15.0 e  | 11.8 e | 14.4    | 10.4 | 46.3      | 26.1  | 20.0    | 34.4  | 14.9      | 25.1  | 280.6          | 0%  |
| 3037.TW                                | Unimicron       | 903.00        | O      | 1465.00      | 44,595     | 14.85 e            | 33.48 e  | 60.8    | 27.0 | 11.0 e  | 8.3 e  | 7.4     | 5.1  | 28.8      | 14.7  | 19.6    | 35.1  | 8.1       | 14.7  | 439.6          | 0%  |
| 6669.TW                                | Wiwynn          | 4,960.00      | O      | 7500.0       | 28,650     | 357.51 e           | 466.82 e | 13.9    | 10.6 | 5.6 e   | 4.3 e  | 0.6     | 0.4  | 9.4       | 7.5   | 46.0    | 45.6  | 17.3      | 16.4  | 242.4          | 3%  |
| HK/ China                              |                 |               |        |              |            |                    |          |         |      |         |        |         |      |           |       |         |       |           |       |                |     |
| 000977.SZ                              | Inspur          | 85.49         | E      | 75.00        | 18,491     | 2.26 e             | 3.03 e   | 37.7    | 28.2 | 4.9 e   | 4.3 e  | 0.6     | 0.4  | 40.2      | 30.5  | 13.9    | 16.3  | 3.7       | 3.7   | 664.3          | 0%  |
| 600183.SS                              | Shengyi         | 134.45        | E      | 74.0         | 48,159     | 2.06 e             | 2.84 e   | 65.2    | 47.3 | 14.9 e  | 12.7 e | 8.3     | 6.6  | 46.8      | 34.7  | 24.6    | 29.1  | 15.7      | 18.3  | 576.6          | 1%  |
| 002916.SZ                              | Shennan Circuit | 393.59        | E      | 400.00       | 39,533     | 7.18 e             | 11.24 e  | 54.8    | 35.0 | 9.3 e   | 7.2 e  | 7.8     | 5.9  | 43.0      | 28.6  | 25.2    | 31.6  | 14.9      | 19.5  | 432.5          | 0%  |
| 0992.HK                                | Lenovo          | 23.30 HKD     | O      | 30.0         | 36,876     | 1.77 e             | 2.22 e   | 13.2    | 10.5 | 3.5 e   | 2.8 e  | 0.4     | 0.3  | 5.5       | 4.4   | 31.6    | 32.0  | 6.1       | 7.0   | 170.9          | 3%  |
| 6088.HK                                | FIT Hong Teng   | 5.99 HKD      | O      | 12.00        | 5,582      | 0.32 e             | 0.53 e   | 18.9    | 11.2 | 1.8 e   | 1.6 e  | 0.9     | 0.8  | 8.9       | 5.4   | 10.2    | 15.1  | 4.7       | 7.3   | 64.4           | 0%  |
| Smartphone & Wireless Equipment        |                 |               |        |              |            |                    |          |         |      |         |        |         |      |           |       |         |       |           |       |                |     |
| Taiwan                                 |                 |               |        |              |            |                    |          |         |      |         |        |         |      |           |       |         |       |           |       |                |     |
| 3008.TW                                | Largan          | 4,345.00      | O      | 4836.00      | 17,664     | 175.24 e           | 182.48 e | 24.8    | 23.8 | 2.7 e   | 2.4 e  | 9.0     | 8.3  | 13.7      | 12.5  | 11.5    | 10.8  | 10.2      | 9.9   | 95.5           | 2%  |
| 3406.TW                                | Genius          | 609.00        | E      | 425.00       | 2,134      | 31.74 e            | 39.14 e  | 19.2    | 15.6 | 2.3 e   | 2.0 e  | 2.7     | 2.5  | 6.6       | 5.6   | 12.7    | 13.7  | 8.1       | 9.4   | 24.2           | 12% |
| HK/ China                              |                 |               |        |              |            |                    |          |         |      |         |        |         |      |           |       |         |       |           |       |                |     |
| 000063.SZ                              | ZTE (A Shr)     | 40.04         | E      | 34.30        | 26,184     | 0.83 e             | 1.16 e   | 48.3    | 34.6 | 2.4 e   | 2.3 e  | 1.2     | 1.0  | 23.4      | 21.3  | 5.2     | 6.9   | 1.8       | 2.3   | 919.4          | 2%  |
| 002036.SZ                              | LCE             | 6.19          | U      | 8.50         | 965        | 0.20 e             | 0.24 e   | 31.0    | 25.4 | 2.2 e   | 2.1 e  | 0.7     | 0.6  | 8.3       | 7.3   | 7.5     | 8.5   | 1.3       | 1.6   | 74.8           | 0%  |
| 002236.SZ                              | Dahua           | 16.79         | E      | 20.80        | 8,094      | 1.35 e             | 1.50 e   | 12.5    | 11.2 | 1.3 e   | 1.3 e  | 1.5     | 1.4  | 9.3       | 8.1   | 11.1    | 11.6  | 8.0       | 8.1   | 141.7          | 4%  |
| 002273.SZ                              | Crystal         | 29.66         | O      | 30.00        | 6,082      | 1.08 e             | 1.25 e   | 27.5    | 23.6 | 4.0 e   | 3.8 e  | 5.1     | 4.4  | 17.8      | 15.3  | 15.0    | 16.7  | 11.9      | 13.0  | 239.9          | 3%  |
| 002373.SZ                              | TransInfo       | 7.03          | E      | 9.70         | 1,428      | 0.44 e             | 0.50 e   | 16.1    | 14.1 | 0.9 e   | 0.8 e  | 1.3     | 1.2  | 6.5       | 5.3   | 5.6     | 6.0   | 3.9       | 4.2   | 57.5           | 0%  |
| 002415.SZ                              | Hikvision       | 33.19         | E      | 35.80        | 44,854     | 1.75 e             | 1.87 e   | 19.0    | 17.7 | 3.1 e   | 2.9 e  | 3.1     | 2.8  | 13.1      | 11.7  | 16.9    | 16.7  | 11.2      | 11.2  | 294.3          | 3%  |
| 002456.SZ                              | SZ O-Film       | 8.79          | U      | 8.00         | 4,531      | 0.06 e             | 0.11 e   | 159.7   | 76.9 | 5.6 e   | 5.1 e  | 1.4     | 1.2  | 27.4      | 22.3  | 3.6     | 6.9   | 0.8       | 1.6   | 187.8          | 1%  |
| 300098.SZ                              | Gosuncn         | 4.28          | U      | 4.80         | 973        | 0.08 e             | 0.10 e   | 57.0    | 42.4 | 2.6 e   | 2.4 e  | 3.1     | 2.6  | 32.5      | 23.6  | 4.6     | 5.9   | 2.5       | 3.0   | 55.4           | 0%  |
| 002281.SZ                              | Accelink        | 216.00        | U      | 166.00       | 26,359     | 2.30 e             | 3.38 e   | 93.9    | 63.9 | 15.0 e  | 12.7 e | 11.1    | 8.2  | 60.6      | 43.4  | 17.1    | 21.6  | 10.6      | 13.1  | 758.3          | 0%  |
| 600745.SS                              | Wingtech        | 21.10         | U      | 15.00        | 3,963      | (0.26) e           | (0.17) e | NM      | NM   | 1.0 e   | 1.1 e  | 7.6     | 7.1  | (550.1)   | 386.3 | (1.3)   | (0.8) | (0.8)     | (0.5) | 238.9          | 0%  |
| 603297.SS                              | Yongxin         | 111.22        | E      | 120.00       | 1,816      | 2.81 e             | 3.37 e   | 39.6    | 33.0 | 5.5 e   | 5.1 e  | 10.6    | 8.7  | 38.6      | 28.8  | 14.5    | 16.0  | 12.3      | 13.6  | 34.2           | 1%  |
| 688036.SS                              | Transsion       | 64.43         | E      | 60.00        | 10,937     | 0.54 e             | 2.14 e   | 118.7   | 30.0 | 3.6 e   | 3.3 e  | 1.1     | 1.0  | 155.3     | 21.4  | 3.0     | 11.5  | 1.4       | 5.1   | 136.0          | 0%  |
| 601869.SS                              | YOFC-A          | 381.87        | U      | 202.25       | 30,395     | 9.07 e             | 15.38 e  | 42.1    | 24.8 | 12.6 e  | 8.9 e  | 7.2     | 5.7  | 19.6      | 12.6  | 35.1    | 42.1  | 18.0      | 23.8  | 511.7          | 0%  |
| 603236.SS                              | Quectel         | 52.30         | E      | 95.00        | 3,108      | 4.25 e             | 4.92 e   | 12.3    | 10.6 | 2.2 e   | 1.8 e  | 0.5     | 0.4  | 8.2       | 7.1   | 19.9    | 18.9  | 6.3       | 6.3   | 113.9          | 0%  |
| 0285.HK                                | BYDE            | 21.46 HKD     | O      | 39.00        | 6,169      | 2.00 e             | 2.63 e   | 10.7    | 8.1  | 1.1 e   | 1.0 e  | 0.2     | 0.2  | 3.8       | 2.8   | 12.2    | 14.5  | 4.7       | 5.7   | 71.0           | 3%  |
| 0763.HK                                | ZTE (H Shr)     | 24.92 HKD     | O      | 39.00        | 26,184     | 0.83 e             | 1.16 e   | 30.1    | 21.6 | 1.3 e   | 1.3 e  | 1.2     | 1.0  | 23.4      | 21.3  | 5.2     | 6.9   | 1.8       | 2.3   | 97.2           | 3%  |
| 1478.HK                                | Q-Tech          | 6.50 HKD      | U      | 7.20         | 991        | 0.72 e             | 0.91 e   | 9.0     | 7.1  | 1.0 e   | 0.9 e  | 0.3     | 0.2  | 5.4       | 3.8   | 12.8    | 14.7  | 4.2       | 4.8   | 9.8            | 4%  |
| 1810.HK                                | Xiaomi          | 25.84 HKD     | O      | 32.00        | 70,359     | 0.89 e             | 1.27 e   | 28.9    | 20.3 | 2.1 e   | 1.9 e  | 1.4     | 1.1  | 28.1      | 16.4  | 5.9     | 8.5   | 3.2       | 4.6   | 852.7          | 0%  |
| 2018.HK                                | AAC             | 38.18 HKD     | O      | 42.00        | 5,838      | 2.06 e             | 2.55 e   | 18.5    | 15.0 | 1.4 e   | 1.3 e  | 1.1     | 1.0  | 5.8       | 4.0   | 9.4     | 10.6  | 4.8       | 5.4   | 33.0           | 1%  |
| 2382.HK                                | Sunny Optical   | 55.05 HKD     | E      | 62.00        | 7,689      | 2.85 e             | 3.46 e   | 19.3    | 15.9 | 1.6 e   | 1.4 e  | 1.2     | 1.0  | 5.8       | 4.2   | 9.9     | 10.7  | 5.0       | 5.4   | 115.9          | 2%  |
| 6869.HK                                | YOFC-H          | 139.70 HKD    | E      | 230.00       | 30,395     | 9.07 e             | 15.38 e  | 15.4    | 9.1  | 4.0 e   | 2.8 e  | 7.2     | 5.7  | 19.6      | 12.6  | 35.1    | 42.1  | 18.0      | 23.8  | 386.9          | 0%  |

NA = not available at this time. Source: FactSet, Morgan Stanley Research; E=Morgan Stanley Research estimates.  
Note: Prices as of the close, July 13, 2026.

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## Hardware Technology Valuation Comparison

| Ticker                          | Company      | Closing Price | Rating | Price Target | Market Cap | EPS (Local Dollar) |          | P/E (X) |         | P/B (X) |        | P/S (X) |      | EV/EBITDA |        | ROE (%) |       | ROA (%) |       | Trading Volume | 26E |
|---------------------------------|--------------|---------------|--------|--------------|------------|--------------------|----------|---------|---------|---------|--------|---------|------|-----------|--------|---------|-------|---------|-------|----------------|-----|
|                                 |              | 7/13/2026     |        | (LC)         | (US\$m)    | 26E                | 27E      | 26E     | 27E     | 26E     | 27E    | 26E     | 27E  | 26E       | 27E    | 26E     | 27E   | 26E     | 27E   | (US\$ m)       |     |
| Tech conglomerate & Enablers    |              |               |        |              |            |                    |          |         |         |         |        |         |      |           |        |         |       |         |       |                |     |
| Taiwan                          |              |               |        |              |            |                    |          |         |         |         |        |         |      |           |        |         |       |         |       |                |     |
| 2301.TW                         | Lite-On Tech | 222.00        | E      | 176.00       | 15,982     | 8.72 e             | 11.19 e  | 25.4    | 19.8    | 5.4 e   | 5.1 e  | 2.4     | 1.9  | 16.48     | 12.66  | 21.56   | 26.28 | 8.99    | 10.08 | 150.49         | 3%  |
| 2308.TW                         | Delta        | 1,890.00      | O      | 2700.0       | 152,592    | 41.52 e            | 65.05 e  | 45.5    | 29.1    | 11.2 e  | 8.7 e  | 6.2     | 4.4  | 25.63     | 17.14  | 29.61   | 36.69 | 14.84   | 18.14 | 528.82         | 1%  |
| 2317.TW                         | Hon Hai      | 236.50        | O      | 310.00       | 103,123    | 16.38 e            | 19.90 e  | 14.4    | 11.9    | 1.6 e   | 1.5 e  | 0.3     | 0.3  | 6.30      | 5.33   | 11.30   | 12.87 | 4.32    | 4.83  | 489.05         | 4%  |
| 2455.TW                         | VPEC         | 329.00        | E      | 290.0        | 1,891      | 4.88 e             | 7.60 e   | 67.4    | 43.3    | 16.4 e  | 13.9 e | 14.6    | 11.5 | 45.75     | 30.82  | 25.73   | 34.64 | 18.26   | 25.06 | 65.48          | 1%  |
| 2474.TW                         | Catcher      | 192.50        | U      | 160.00       | 3,224      | 12.03 e            | 12.52 e  | 16.0    | 15.4    | 0.7 e   | 0.7 e  | 6.1     | 5.7  | 27.17     | 26.27  | 4.65    | 4.81  | 3.23    | 3.32  | 27.89          | 6%  |
| 3017.TW                         | AVC          | 2,225.00      | O      | 3333.0       | 27,147     | 96.50 e            | 145.03 e | 23.1    | 15.3    | 11.0 e  | 7.2 e  | 3.9     | 2.9  | 13.21     | 8.72   | 57.33   | 55.61 | 19.71   | 22.07 | 325.10         | 1%  |
| 3081.TWO                        | LandMark     | 1,990.00      | E      | 2850.00      | 5,722      | 17.62 e            | 28.78 e  | 113.0   | 69.1    | 30.3 e  | 18.2 e | 39.3    | 24.4 | 80.48     | 50.83  | 31.52   | 32.71 | 26.72   | 28.67 | 108.59         | 0%  |
| 3324.TWO                        | Auras        | 905.00        | E      | 1025.0       | 2,618      | 58.09 e            | 66.06 e  | 15.6    | 13.7    | 5.3 e   | 4.3 e  | 2.0     | 1.8  | 10.57     | 9.12   | 39.64   | 34.72 | 17.35   | 15.65 | 124.41         | 1%  |
| 6805.TW                         | Fositek      | 1,410.00      | O      | 2426.00      | 3,004      | 67.54 e            | 92.07 e  | 20.9    | 15.3    | 8.7 e   | 6.1 e  | 5.0     | 3.9  | 14.97     | 10.52  | 50.61   | 46.84 | 25.85   | 25.70 | 107.39         | 1%  |
| HK/ China                       |              |               |        |              |            |                    |          |         |         |         |        |         |      |           |        |         |       |         |       |                |     |
| 002241.SZ                       | GoerTek      | 21.82         | U      | 20.50        | 10,113     | 1.29 e             | 1.38 e   | 17.0    | 15.8    | 2.0 e   | 1.9 e  | 0.7     | 0.7  | 9.18      | 7.69   | 12.04   | 12.13 | 4.92    | 4.76  | 452.99         | 3%  |
| 002475.SZ                       | Luxshare     | 58.98         | O      | 77.00        | 66,711     | 2.85 e             | 4.05 e   | 20.7    | 14.6    | 3.5 e   | 2.8 e  | 1.0     | 0.9  | 11.75     | 9.10   | 18.48   | 21.51 | 7.72    | 9.53  | 1,361.96       | 0%  |
| 002600.SZ                       | Lingyi       | 13.64         | U      | 12.20        | 15,188     | 0.51 e             | 0.67 e   | 27.0    | 20.3    | 0.9 e   | 0.8 e  | 1.5     | 1.2  | 12.74     | 10.59  | 13.35   | 15.34 | 5.94    | 7.29  | 497.14         | 0%  |
| 601138.SS                       | FII          | 63.08         | O      | 82.8         | 184,583    | 3.08 e             | 4.04 e   | 20.5    | 15.6    | 7.3 e   | 7.0 e  | 0.9     | 0.7  | 14.13     | 11.23  | 36.07   | 45.92 | 11.11   | 11.57 | 1,717.18       | 5%  |
| Display & Industrial Automation |              |               |        |              |            |                    |          |         |         |         |        |         |      |           |        |         |       |         |       |                |     |
| Taiwan                          |              |               |        |              |            |                    |          |         |         |         |        |         |      |           |        |         |       |         |       |                |     |
| 1590.TW                         | Airtac       | 1,320.00      | O      | 2000.0       | 8,206      | 56.77 e            | 65.77 e  | 23.3    | 20.1    | 4.5 e   | 4.0 e  | 6.2     | 5.4  | 15.70     | 13.51  | 20.53   | 21.28 | 18.56   | 19.04 | 30.71          | 2%  |
| 2049.TW                         | Hiwin        | 332.00        | O      | 400.0        | 3,279      | 8.77 e             | 10.77 e  | 37.9    | 30.8    | 2.2 e   | 2.0 e  | 3.9     | 3.4  | 19.30     | 16.40  | 8.02    | 9.22  | 5.36    | 5.88  | 38.49          | 1%  |
| 2360.TW                         | Chroma       | 1,850.00      | O      | 2800.0       | 24,452     | 39.87 e            | 55.57 e  | 46.4    | 33.3    | 18.8 e  | 14.5 e | 15.7    | 11.8 | 36.82     | 26.58  | 45.59   | 49.12 | 30.31   | 32.01 | 140.03         | 1%  |
| 2395.TW                         | Advantech    | 536.00        | O      | 400.0        | 14,465     | 14.77 e            | 16.97 e  | 36.3    | 31.6    | 7.8 e   | 7.3 e  | 5.7     | 5.1  | 29.59     | 25.43  | 22.18   | 23.85 | 15.32   | 16.27 | 31.99          | 2%  |
| 2409.TW                         | AJO          | 29.25         | E      | 27.0         | 6,861      | (0.13) e           | 0.21 e   | NM      | 137.8   | 1.5 e   | 1.5 e  | 0.8     | 0.8  | 6.42      | 4.63   | (0.67)  | 1.08  | (0.24)  | 0.37  | 104.73         | 0%  |
| 3481.TW                         | Innolux      | 61.00         | E      | 60.0         | 15,148     | 1.57 e             | 1.67 e   | 38.9    | 36.5    | 2.1 e   | 2.0 e  | 1.8     | 1.7  | 10.80     | 9.49   | 5.50    | 5.66  | 3.21    | 3.33  | 240.00         | 1%  |
| 3665.TW                         | Bizlink      | 1,865.00      | O      | 3665.0       | 11,308     | 63.77 e            | 120.77 e | 29.2    | 15.4    | 6.5 e   | 4.8 e  | 3.7     | 2.4  | 18.08     | 10.06  | 24.01   | 35.51 | 13.85   | 21.31 | 180.84         | 1%  |
| 3714.TW                         | Ennostar     | 60.90         | U      | 29.0         | 1,397      | (0.08) e           | 0.01 e   | NM      | 6,803.7 | 0.9 e   | 0.9 e  | 1.8     | 1.7  | 7.13      | 8.77   | (0.11)  | 0.01  | (0.09)  | 0.01  | 16.37          | 0%  |
| 6176.TW                         | Radiant      | 90.00         | E      | 170.0        | 1,301      | 11.47 e            | 11.27 e  | 7.8     | 8.0     | 1.1 e   | 1.0 e  | 0.9     | 0.9  | 0.26      | 0.05   | 13.99   | 13.23 | 7.40    | 7.20  | 15.31          | 9%  |
| 8069.TWO                        | E Ink        | 196.00        | O      | 250.0        | 7,032      | 11.37 e            | 13.07 e  | 17.2    | 15.0    | 3.0 e   | 2.8 e  | 5.2     | 4.5  | 13.47     | 11.35  | 17.91   | 18.95 | 11.82   | 12.80 | 42.25          | 4%  |
| 2059.TW                         | King Slide   | 8,080.00      | O      | 6650.0       | 23,933     | 167.77 e           | 193.77 e | 48.2    | 41.7    | 20.0 e  | 15.0 e | 28.5    | 24.1 | 38.36     | 31.65  | 48.03   | 41.12 | 38.12   | 32.93 | 100.12         | 1%  |
| 2345.TW                         | Accton       | 2,405.00      | O      | 3800.0       | 41,945     | 85.77 e            | 115.77 e | 28.0    | 20.8    | 14.2 e  | 9.3 e  | 3.4     | 2.7  | 19.82     | 14.18  | 63.07   | 54.29 | 26.93   | 25.91 | 231.33         | 1%  |
| 8210.TW                         | Chenbro      | 1,135.00      | O      | 1780.0       | 4,421      | 47.77 e            | 58.77 e  | 23.8    | 19.3    | 9.4 e   | 7.3 e  | 4.2     | 3.3  | 16.10     | 12.88  | 44.50   | 41.35 | 21.17   | 20.25 | 72.34          | 1%  |
| HK/ China                       |              |               |        |              |            |                    |          |         |         |         |        |         |      |           |        |         |       |         |       |                |     |
| 000050.SZ                       | Tianma       | 7.16          | U      | 6.3          | 2,508      | 0.21 e             | 0.25 e   | 33.4    | 28.5    | 0.6 e   | 0.6 e  | 0.4     | 0.4  | 4.91      | 4.22   | 1.77    | 2.04  | 0.66    | 0.74  | 50.25          | 0%  |
| 000100.SZ                       | TCL Corp.    | 4.81          | E      | 4.7          | 14,753     | 0.32 e             | 0.37 e   | 15.3    | 12.9    | 1.7 e   | 1.8 e  | 0.4     | 0.4  | 4.94      | 4.61   | 11.45   | 13.54 | 1.48    | 1.70  | 372.40         | 4%  |
| 000725.SZ                       | BOE Tech     | 6.83          | O      | 9.3          | 37,025     | 0.20 e             | 0.27 e   | 35.0    | 25.7    | 1.8 e   | 1.7 e  | 1.2     | 1.2  | 6.04      | 5.93   | 5.33    | 6.92  | 1.59    | 1.98  | 732.44         | 1%  |
| 300433.SZ                       | Lens Tech    | 39.53         | E      | 31.0         | 29,832     | 0.82 e             | 1.39 e   | 48.0    | 28.4    | 3.7 e   | 3.3 e  | 2.6     | 2.0  | 16.68     | 12.71  | 7.47    | 11.75 | 4.85    | 7.47  | 536.38         | 1%  |
| 300567.SZ                       | Jingce       | 297.10        | E      | 70.0         | 12,256     | 0.87 e             | 1.47 e   | 342.9   | 202.5   | 19.1 e  | 17.8 e | 25.1    | 22.0 | 276.31    | 162.72 | 5.73    | 9.12  | 2.45    | 3.85  | 177.30         | 0%  |

NA = not available at this time. Source: FactSet, Morgan Stanley Research; E=Morgan Stanley Research estimates.  
Note: Prices as of the close, July 13, 2026.

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## Demand Forecasts – Potential Volume Downside from Memory Cost Hike

| (mn units)                        | 2008        | 2009        | 2010        | 2011        | 2012        | 2013         | 2014         | 2015         | 2016         | 2017         | 2018         | 2019         | 2020         | 2021         | 2022         | 2023         | 2024         | 2025         | 2026e        | 2027e        |
|-----------------------------------|-------------|-------------|-------------|-------------|-------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| <b>Server</b>                     | <b>8.4</b>  | <b>7.1</b>  | <b>8.3</b>  | <b>8.6</b>  | <b>8.7</b>  | <b>9.0</b>   | <b>9.2</b>   | <b>9.7</b>   | <b>9.6</b>   | <b>10.2</b>  | <b>11.8</b>  | <b>11.8</b>  | <b>12.7</b>  | <b>13.5</b>  | <b>14.9</b>  | <b>12.1</b>  | <b>13.8</b>  | <b>16.4</b>  | <b>20.1</b>  | <b>22.1</b>  |
| YoY % Change                      |             | -16%        | 17%         | 4%          | 0%          | 3%           | 3%           | 5%           | -2%          | 7%           | 16%          | 0%           | 7%           | 7%           | 10%          | -19%         | 14%          | 19%          | 22%          | 10%          |
| <b>PC</b>                         | <b>297</b>  | <b>312</b>  | <b>355</b>  | <b>360</b>  | <b>346</b>  | <b>312</b>   | <b>305</b>   | <b>272</b>   | <b>256</b>   | <b>261</b>   | <b>260</b>   | <b>268</b>   | <b>307</b>   | <b>354</b>   | <b>294</b>   | <b>253</b>   | <b>262</b>   | <b>289</b>   | <b>250</b>   | <b>241</b>   |
| YoY % Change                      |             | 11%         | 5%          | 14%         | 2%          | -4%          | -10%         | -2%          | -11%         | -6%          | 2%           | 0%           | 3%           | 15%          | 15%          | -17%         | -14%         | 3%           | 10%          | -13%         |
| <b>Desktop</b>                    | <b>155</b>  | <b>144</b>  | <b>155</b>  | <b>152</b>  | <b>146</b>  | <b>134</b>   | <b>131</b>   | <b>111</b>   | <b>101</b>   | <b>101</b>   | <b>96</b>    | <b>96</b>    | <b>86</b>    | <b>96</b>    | <b>86</b>    | <b>72</b>    | <b>73</b>    | <b>84</b>    | <b>73</b>    | <b>72</b>    |
| YoY % Change                      |             | -4%         | -7%         | 7%          | -2%         | -4%          | -8%          | -2%          | -16%         | -9%          | 0%           | -5%          | 0%           | -11%         | 12%          | -11%         | -16%         | 1%           | 15%          | -12%         |
| <b>Notebook</b>                   | <b>142</b>  | <b>168</b>  | <b>200</b>  | <b>208</b>  | <b>200</b>  | <b>178</b>   | <b>173</b>   | <b>161</b>   | <b>155</b>   | <b>161</b>   | <b>164</b>   | <b>172</b>   | <b>221</b>   | <b>258</b>   | <b>208</b>   | <b>181</b>   | <b>189</b>   | <b>205</b>   | <b>177</b>   | <b>170</b>   |
| YoY % Change                      |             | 33%         | 19%         | 19%         | 4%          | -4%          | -11%         | -2%          | -7%          | -4%          | 3%           | 2%           | 5%           | 29%          | 17%          | -19%         | -13%         | 4%           | 9%           | -14%         |
| <b>Tablet</b>                     |             |             | <b>19</b>   | <b>76</b>   | <b>144</b>  | <b>220</b>   | <b>230</b>   | <b>207</b>   | <b>175</b>   | <b>164</b>   | <b>146</b>   | <b>144</b>   | <b>164</b>   | <b>168</b>   | <b>162</b>   | <b>129</b>   | <b>145</b>   | <b>148</b>   | <b>151</b>   | <b>154</b>   |
| YoY % Change                      |             |             |             | 294%        | 89%         | 52%          | 5%           | -10%         | -16%         | -6%          | -11%         | -1%          | 13%          | 3%           | -4%          | -20%         | 12%          | 2%           | 2%           | 2%           |
| <b>Smartphone</b>                 | <b>151</b>  | <b>173</b>  | <b>305</b>  | <b>494</b>  | <b>727</b>  | <b>1,019</b> | <b>1,301</b> | <b>1,438</b> | <b>1,469</b> | <b>1,465</b> | <b>1,403</b> | <b>1,373</b> | <b>1,281</b> | <b>1,360</b> | <b>1,206</b> | <b>1,165</b> | <b>1,237</b> | <b>1,264</b> | <b>1,094</b> | <b>1,133</b> |
| YoY % Change                      |             | 22%         | 15%         | 76%         | 62%         | 47%          | 40%          | 28%          | 10%          | 2%           | 0%           | -4%          | -2%          | -7%          | 6%           | -11%         | -3%          | 6%           | 2%           | -13%         |
| <b>iPhone</b>                     | <b>14</b>   | <b>25</b>   | <b>47</b>   | <b>93</b>   | <b>136</b>  | <b>153</b>   | <b>193</b>   | <b>232</b>   | <b>215</b>   | <b>216</b>   | <b>205</b>   | <b>199</b>   | <b>203</b>   | <b>236</b>   | <b>227</b>   | <b>236</b>   | <b>234</b>   | <b>252</b>   | <b>253</b>   | <b>267</b>   |
| YoY % Change                      |             | 269%        | 84%         | 89%         | 96%         | 46%          | 13%          | 26%          | 20%          | -7%          | 0%           | -5%          | -3%          | 2%           | 16%          | -4%          | 4%           | -1%          | 7%           | 0%           |
| <b>iPad</b>                       |             |             | <b>15</b>   | <b>40</b>   | <b>66</b>   | <b>74</b>    | <b>63</b>    | <b>50</b>    | <b>43</b>    | <b>44</b>    | <b>43</b>    | <b>50</b>    | <b>53</b>    | <b>58</b>    | <b>60</b>    | <b>49</b>    | <b>51</b>    | <b>54</b>    | <b>56</b>    | <b>57</b>    |
| YoY % Change                      |             |             |             | 174%        | 62%         | 13%          | -15%         | -22%         | -14%         | 3%           | -1%          | 15%          | 7%           | 8%           | 5%           | -20%         | 4%           | 8%           | 2%           | 1%           |
| <b>Global Light Vehicle Sales</b> | <b>56.5</b> | <b>54.1</b> | <b>60.7</b> | <b>63.9</b> | <b>67.6</b> | <b>70.4</b>  | <b>74.0</b>  | <b>76.2</b>  | <b>80.2</b>  | <b>82.3</b>  | <b>81.7</b>  | <b>78.1</b>  | <b>66.9</b>  | <b>69.0</b>  | <b>68.5</b>  | <b>75.4</b>  | <b>74.7</b>  | <b>77.2</b>  | <b>ok</b>    |              |
| YoY % Change                      |             | -6%         | -4%         | 12%         | 5%          | 6%           | 4%           | 5%           | 3%           | 5%           | 3%           | -1%          | -4%          | -14%         | 3%           | -1%          | 10%          | -1%          | 3%           |              |

Source: IDC, Morgan Stanley Research. e = Morgan Stanley Research estimates.

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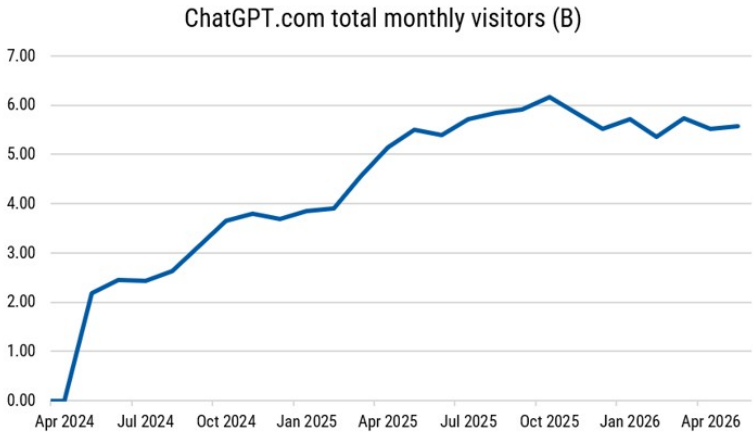
## Valuation Premium Justified for the AI Hardware Supply Chain

| Company      | Ticker    | AI revenue mix |        | AI profit mix |        | P/E   |       | PEG   |        |
|--------------|-----------|----------------|--------|---------------|--------|-------|-------|-------|--------|
|              |           | 2025e          | 2026e  | 2025e         | 2026e  | 2026e | 2027e | 2026e | 2027e  |
| Wistron      | 3231.TW   | 27%            | 35%    | 6%            | 26%    | 10.3x | 7.9x  | 0.19x | 0.27x  |
| Bizlink      | 3665.TW   | 15-20%         | 20-25% | 20-25%        | 25-30% | 29.2x | 15.4x | 0.79x | 0.17x  |
| FII          | 601138.SS | 52%            | 56%    | 32%           | 35%    | 20.5x | 15.6x | 0.28x | 0.50x  |
| Lite-on Tech | 2301.TW   | 12%            | 11%    | 14%           | 13%    | 25.4x | 19.8x | 0.81x | 0.70x  |
| Delta        | 2308.TW   | 15%            | 20%    | 16%           | 22%    | 45.5x | 29.1x | 0.57x | 0.51x  |
| AVC          | 3017.TW   | 15%            | 17%    | 15%           | 17%    | 23.1x | 15.3x | 0.24x | 0.31x  |
| Accton       | 2345.TW   | 70-75%         | 75-80% | 70-75%        | 75-80% | 28.0x | 20.8x | 0.34x | 0.59x  |
| Auras        | 3324.TWO  | 38%            | 47%    | 50%           | 62%    | 15.6x | 13.7x | 0.15x | 1.00x  |
| King Slide   | 2059.TW   | 55%            | 55-60% | 60%           | 60-65% | 48.2x | 41.7x | 0.77x | 2.69x  |
| Accelink     | 002281.SZ | 0-5%           | 20%    | 0-5%          | 30%    | 93.9x | 63.9x | 0.98x | 1.36x  |
| Quanta       | 2382.TW   | 50%            | 68%    | 35%           | 56%    | 15.3x | 13.1x | 0.57x | 0.78x  |
| Giga-Byte    | 2376.TW   | 54%            | 61%    | 26%           | 30%    | 10.4x | 11.3x | 0.13x | -1.38x |

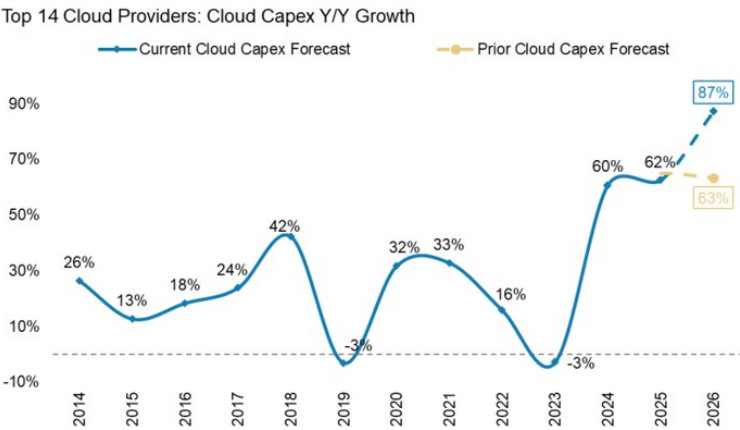
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates. Note: Based on closing prices as of July 13, 2026.

AI Server Momentum Continues To Be Strong

ChatGPT Monthly Visitors (B)

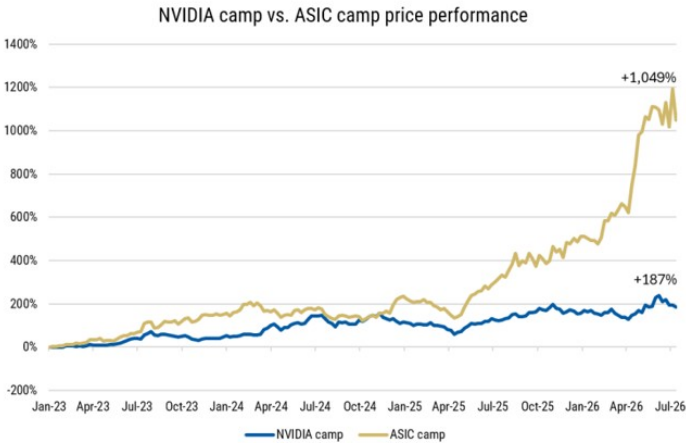


Total Cloud Capex Up Y/Y in CY25-26

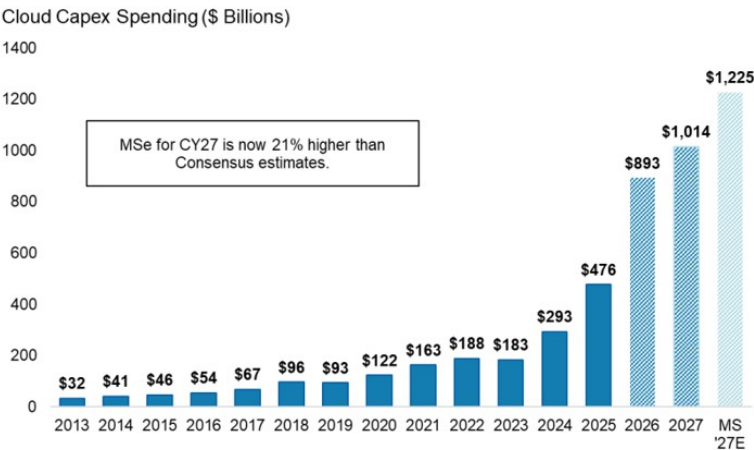


Source: IDC, Morgan Stanley Research. Morgan Stanley Research estimates.

GPU vs. ASIC Share Price Movement



Cloud Capex Could See Upside for CY26



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## Agentic AI – Hardware Beneficiaries (ABF Substrates, CPU Sockets, MLCCs)

| Every 1M Units of total CPU upside               |             | If CPU TAM grows 10-15M units...                                   |             |
|--------------------------------------------------|-------------|--------------------------------------------------------------------|-------------|
| Server CPU ABF Substrate ASP (US\$)              | 77.8        | Server CPU ABF Substrate ASP (US\$)                                | 77.8        |
| Revenue upside to Unimicron per 1M CPUs (US\$ M) | 12.6        | Revenue upside to Unimicron if CPU TAM grows 10-15M units (US\$ M) | 157.6       |
| Revenue upside to Unimicron per 1M CPUs (NT\$ M) | 397         | Revenue upside to Unimicron if CPU TAM grows 10-15M units (NT\$ M) | 4,964       |
| Unimicron 2026 total revenue estimate (NT\$ M)   | 169,246     | Unimicron 2026 ABF revenue estimate (NT\$ M)                       | 169,246     |
| <b>Unimicron ABF revenue upside per 1M CPUs</b>  | <b>0.2%</b> | <b>Unimicron ABF revenue upside if CPU TAM grows 10-15M units</b>  | <b>2.9%</b> |

| Every 1M Units of total CPU upside           |             | If CPU TAM grows 10-15M units...                               |             |
|----------------------------------------------|-------------|----------------------------------------------------------------|-------------|
| Server CPU Socket ASP (US\$)                 | 21.9        | Server CPU Socket ASP (US\$)                                   | 21.9        |
| Revenue upside to Lotes per 1M CPUs (US\$ M) | 7.7         | Revenue upside to Lotes if CPU TAM grows 10-15M units (US\$ M) | 95.7        |
| Revenue upside to Lotes per 1M CPUs (NT\$ M) | 241         | Revenue upside to Lotes if CPU TAM grows 10-15M units (NT\$ M) | 3,015       |
| Lotes 2026 total revenue estimate (NT\$ M)   | 41,065      | Lotes 2026 total revenue estimate (NT\$ M)                     | 41,065      |
| <b>Lotes CPU Socket upside per 1M CPUs</b>   | <b>0.6%</b> | <b>Lotes CPU Socket upside if CPU TAM grows 10-15M units</b>   | <b>7.3%</b> |

| Every 1M Units of total CPU upside         |             | If CPU TAM grows 10-15M units...                             |             |
|--------------------------------------------|-------------|--------------------------------------------------------------|-------------|
| Server CPU Socket ASP (US\$)               | 21.9        | Server CPU Socket ASP (US\$)                                 | 21.9        |
| Revenue upside to FIT per 1M CPUs (US\$ M) | 8.8         | Revenue upside to FIT if CPU TAM grows 10-15M units (US\$ M) | 109.4       |
| FIT 2026 total revenue estimate (US\$ M)   | 5,483       | FIT 2026 ABF revenue estimate (NT\$ M)                       | 5,483       |
| <b>FIT CPU Socket upside per 1M CPUs</b>   | <b>0.2%</b> | <b>FIT CPU Socket upside if CPU TAM grows 10-15M units</b>   | <b>2.0%</b> |

| Every 1M Units of total CPU upside           |             | If CPU TAM grows 10-15M units...                               |             |
|----------------------------------------------|-------------|----------------------------------------------------------------|-------------|
| Server MLCC Content (US\$)                   | 30.0        | Server MLCC Content (US\$)                                     | 30.0        |
| Revenue upside to Yageo per 1M CPUs (US\$ M) | 4.5         | Revenue upside to Yageo if CPU TAM grows 10-15M units (US\$ M) | 56.3        |
| Revenue upside to Yageo per 1M CPUs (NT\$ M) | 142         | Revenue upside to Yageo if CPU TAM grows 10-15M units (NT\$ M) | 1,772       |
| Yageo 2026 total revenue estimate (NT\$ M)   | 158,893     | Yageo 2026 total revenue estimate (NT\$ M)                     | 158,893     |
| <b>Yageo MLCC revenue upside per 1M CPUs</b> | <b>0.1%</b> | <b>Yageo MLCC revenue upside if CPU TAM grows 10-15M units</b> | <b>1.1%</b> |

Source: Morgan Stanley Research estimates.



## NVIDIA GPU Roadmap

### Recent updates

- 2025 came in at ~29K racks, and we forecast 70-80K racks for 2026.
- VR200 rack production is delayed by 1-2 months vs original schedule due to component issues (i.e., Midplane PCB), so rack ramp will start in 4Q26.
- Quanta should be a share gainer for GB racks in CY26.

| NVIDIA              | H100                   | H200                | B200                 | GB200 NVL72                                     | GB300 NVL72<br>(RENAMED FROM GB200 Ultra)       | VR200 NVL144                                    | VR200 NVL144 CPX                                | VR300                |
|---------------------|------------------------|---------------------|----------------------|-------------------------------------------------|-------------------------------------------------|-------------------------------------------------|-------------------------------------------------|----------------------|
| Launch timing       | 4Q22                   | 2Q24                | 1Q25                 | 4Q24                                            | 3Q25                                            | 2H26                                            | End of 2026                                     | 2H27                 |
| CPU                 | x86                    | x86                 | x86                  | Grace                                           | Grace                                           | Vera                                            | Vera                                            | Vera                 |
| Back-end            | CoWoS-S                | CoWoS-S             | CoWoS-L              | CoWoS-L                                         | CoWoS-L                                         | CoWoS-L                                         | CoWoS-L                                         | CoWoS-L              |
| GPU Cooling         | Air                    | Air & Liquid        | Air & Liquid         | Liquid                                          | Liquid                                          | Liquid                                          | Liquid                                          | Liquid               |
| GPU Max TDP         | 700W                   | 700W                | 1,000W               | 1,200W                                          | 1,400W                                          | 2,300W                                          | 800W                                            | 3,600W               |
| Product Form Factor | DGX/HGX 8x GPU PCIe    | DGX/HGX 8x GPU PCIe | DGX/HGX 8x GPU PCIe  | Oberon GB 2:4 board NVSwitch tray NVL backplane | Oberon GB 2:4 board NVSwitch tray NVL backplane | Oberon VR 2:4 board NVSwitch tray NVL backplane | Oberon VR 2:4 board NVSwitch tray NVL backplane | Kyber                |
| GPU Memory          | 80/96GB   HBM2e/3   8H | 141GB   HBM3e   8H  | 192GB   HBM3e   8H   | 192GB   HBM3e   8H                              | 288GB   HBM3e   8H                              | 288GB   HBM4   8H                               | 128GB GDDR7                                     | 1TB   HBM4e   16H    |
| HBM Vendor          | Hynix                  | Hynix/Micron        | Hynix/Micron/Samsung | Hynix/Micron/Samsung                            | Hynix/Micron/Samsung                            | Hynix/Micron/Samsung                            | Hynix/Micron/Samsung                            | Hynix/Micron/Samsung |
| NVLink B/W per GPU  | 900GB/s                | 900GB/s             | 1,800 GB/s           | 1,800 GB/s                                      | 1,800 GB/s                                      | 3,600 GB/s                                      | 3,600 GB/s                                      | 3,600 GB/s           |
| CPU-GPU Interface   | 128 GB/s PCIe5         | 128 GB/s PCIe5      | 128 GB/s PCIe5       | 450 GB/s C2C                                    | 450 GB/s C2C                                    | 1.8TB/s C2C                                     | 1.8TB/s C2C                                     | 1.8TB/s C2C          |
| Network Adapter     | 400G CX7   BF3         | 400G CX7   BF3      | 400G CX7   BF3       | 400G CX7   BF3                                  | 800G CX8                                        | 1,600G CX9                                      | 1,600G CX9                                      | 1,600G CX9           |

Source: Morgan Stanley Research.

Morgan Stanley

## AMD GPU Roadmap

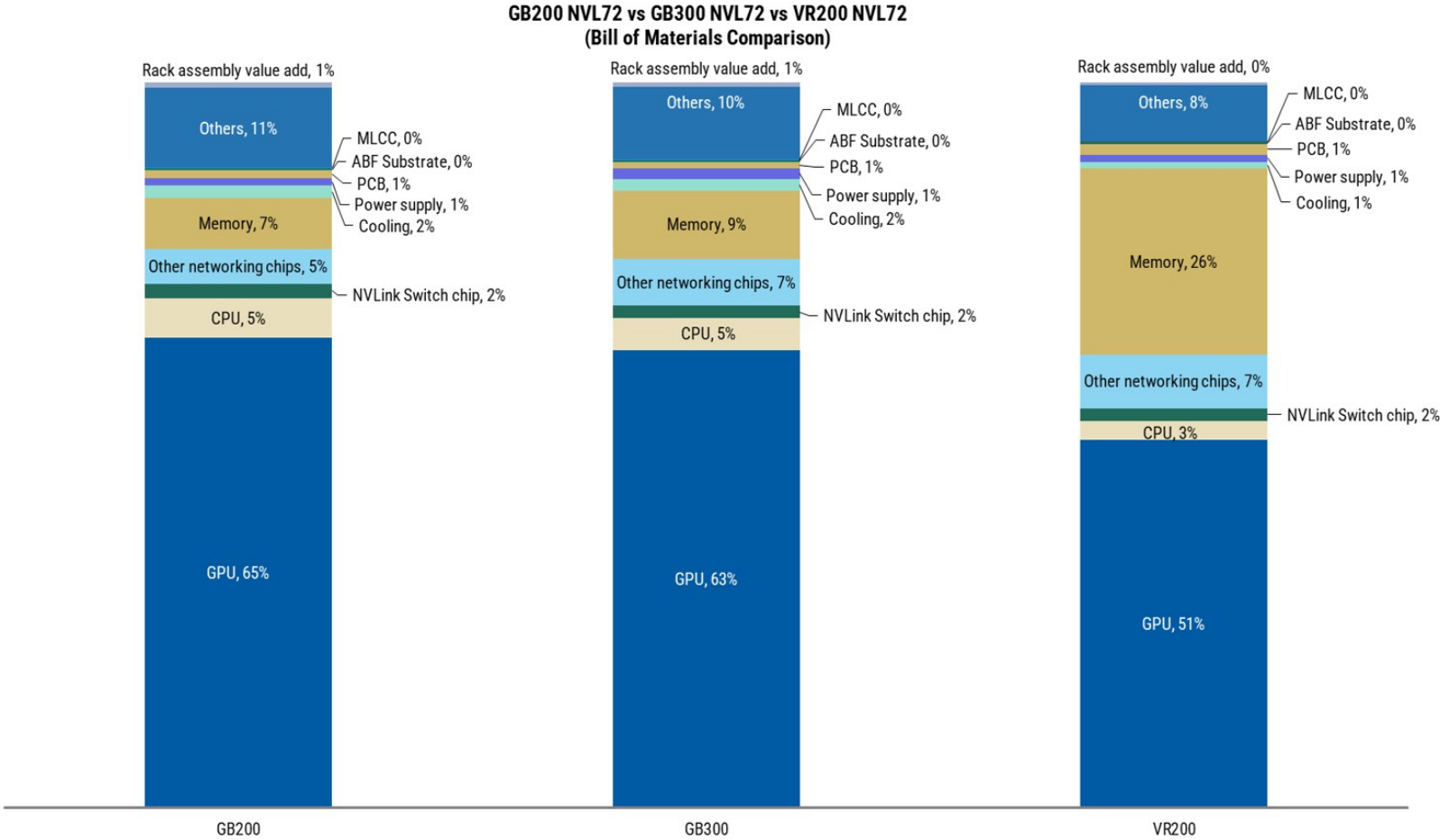
### Recent updates

- MI400-series Helios rack OAM/UBB are produced by Wistron.
- Wiwynn will be the main ODM partner for Meta for MI400-series Helios rack, and should be the main beneficiary of Meta/AMD's multi-year, multi-generation partnership to deploy up to 6 gigawatts of AMD Instinct GPUs.

| AMD                 | MI250X         | MI300A         | MI300X         | MI325X         | MI350X         |
|---------------------|----------------|----------------|----------------|----------------|----------------|
| Launch timing       | 4Q21           | 4Q23           | 4Q23           | 4Q24           | 3Q25           |
| CPU                 | x86            | x86            | x86            | x86            | x86            |
| Back-end            | SolC + CoWoS-S | SolC + CoWoS-S | SolC + CoWoS-S | SolC + CoWoS-S | SolC + CoWoS-S |
| GPU Cooling         | Air            | Air            | Air            | Air            | Air            |
| GPU Max TDP         | 560W           | 760W           | 750W           | 750W           | 750W           |
| Product Form Factor | 4x GPU         | 4x APU         | 8x GPU         | 8x GPU         | N/A            |
| GPU Memory          | 128GB   HBM2e  | 128GB   HBM3   | 192GB   HBM3   | 288GB   HBM3e  | 288GB   HBM3e  |
| HBM Vendor          | Samsung        | Samsung        | Samsung        | N/A            | N/A            |
| B/W per GPU         | 200GB/s        | 384GB/s        | 896GB/s        | N/A            | N/A            |
| CPU-GPU Interface   | 64 GB/s PCIe4  | N/A            | 128 GB/s PCIe5 | 128 GB/s PCIe5 | 128 GB/s PCIe5 |

Source: Morgan Stanley Research estimates.

Owing to recent increase in memory prices, memory will become 25%+ of Rubin BOM



Source: Morgan Stanley Research estimates.

Morgan Stanley

**We estimate that a single VR200 NVL72 rack will cost ~US\$7.8M for hyperscalers**

| <b>Nvidia NVL72 Bill of Materials</b> | <b>GB300</b>       | <b>VR200</b>       | <b>Diff.</b> |
|---------------------------------------|--------------------|--------------------|--------------|
| GPU                                   | \$2,520,000        | \$3,960,000        | 57%          |
| CPU                                   | \$180,000          | \$180,000          | 0%           |
| NVLink Switch chip                    | \$64,800           | \$144,000          | 122%         |
| Other networking chips                | \$261,000          | \$576,000          | 121%         |
| Memory                                | \$373,939          | \$2,001,600        | 435%         |
| Cooling                               | \$64,610           | \$72,080           | 12%          |
| Power supply                          | \$57,600           | \$76,000           | 32%          |
| PCB                                   | \$35,100           | \$116,730          | 233%         |
| ABF Substrate                         | \$11,160           | \$20,340           | 82%          |
| MLCC                                  | \$1,530            | \$4,320            | 182%         |
| Others                                | \$402,412          | \$623,278          | 55%          |
| Rack assembly value add               | \$22,400           | \$28,800           | 29%          |
| <b>Total</b>                          | <b>\$3,994,551</b> | <b>\$7,803,148</b> | <b>95%</b>   |

Source: Morgan Stanley Research estimates.

Morgan Stanley

**Our bottom-up analysis shows ODM value-added per rack to increase ~38% for VR200**

| <b>ODM Value-Add per rack</b>  | <b>GB300</b>     | <b>VR200</b>     |
|--------------------------------|------------------|------------------|
| Compute Board Assembly/Test    | \$12,096         | \$16,200         |
| Compute Tray Assembly/Test     | \$28,800         | \$32,400         |
| Switch Board Assembly/Test     | \$2,475          | \$3,150          |
| Switch Tray Assembly/Test      | \$2,700          | \$3,150          |
| Rack Assembly/Test             | \$22,400         | \$28,800         |
| BF DPU Assembly/Test           | \$1,170          | \$1,170          |
| CX/Orchid Module Assembly/Test | \$0              | \$3,600          |
| Other peripheral boards        | \$3,150          | \$1,260          |
| Cooling components             | \$12,922         | \$14,416         |
| Others                         | \$22,500         | \$45,500         |
| <b>Total ODM Value-Add</b>     | <b>\$108,213</b> | <b>\$149,646</b> |

Source: Morgan Stanley Research estimates.

Morgan Stanley

## Our bottom-up analysis shows PCB content increasing ~233% for VR200

| PCB ASP per board (US\$)          | GB300           | VR200            |
|-----------------------------------|-----------------|------------------|
| Compute PCB                       | \$650           | \$1,400          |
| Switch PCB                        | \$800           | \$1,450          |
| Midplane PCB                      | \$0             | \$1,500          |
| BlueField PCB                     | \$0             | \$255            |
| ConnectX PCB                      | \$0             | \$270            |
| Other peripheral PCB              | \$50            | \$50             |
| PCB Units per rack                | GB300           | VR200            |
| Compute PCB                       | 36x             | 36x              |
| Switch PCB                        | 9x              | 9x               |
| Midplane PCB                      | 0x              | 18x              |
| BlueField PCB                     | 18x             | 18x              |
| ConnectX PCB                      | 0x              | 72x              |
| Other peripheral PCB              | 90x             | 45x              |
| Total PCB Content per rack        | GB300           | VR200            |
| Compute PCB                       | \$23,400        | \$50,400         |
| Switch PCB                        | \$7,200         | \$13,050         |
| Midplane PCB                      | \$0             | \$27,000         |
| BlueField PCB                     | \$0             | \$4,590          |
| ConnectX PCB                      | \$0             | \$19,440         |
| Other peripheral PCB              | \$4,500         | \$2,250          |
| <b>Total PCB Content per rack</b> | <b>\$35,100</b> | <b>\$116,730</b> |

Source: Morgan Stanley Research estimates.



Morgan Stanley

## Our bottom-up analysis shows MLCC content increasing ~182% for VR200

| <b>MLCC content per board (US\$)</b> | <b>GB300</b>   | <b>VR200</b>   |
|--------------------------------------|----------------|----------------|
| Compute PCB                          | \$25           | \$90           |
| Switch PCB                           | \$20           | \$45           |
| BlueField DPU Module                 | \$5            | \$5            |
| ConnectX Orchid Module               | \$5            | \$5            |
| Other peripheral PCB                 | \$5            | \$5            |
| <b>Units per rack</b>                | <b>GB300</b>   | <b>VR200</b>   |
| Compute PCB                          | 36x            | 36x            |
| Switch PCB                           | 9x             | 9x             |
| BlueField DPU Module                 | 0x             | 18x            |
| ConnectX Orchid Module               | 0x             | 72x            |
| Other peripheral PCB                 | 90x            | 45x            |
| <b>Total MLCC Content per rack</b>   | <b>GB300</b>   | <b>VR200</b>   |
| Compute PCB                          | \$900          | \$3,240        |
| Switch PCB                           | \$180          | \$405          |
| BlueField DPU Module                 | \$0            | \$90           |
| ConnectX Orchid Module               | \$0            | \$360          |
| Other peripheral PCB                 | \$450          | \$225          |
| <b>Total MLCC Content per rack</b>   | <b>\$1,530</b> | <b>\$4,320</b> |

Source: Morgan Stanley Research estimates.

Morgan Stanley

## Our bottom-up analysis shows ABF substrate content increasing ~82% for VR200

| ABF Substrate ASP per chip (US\$) | GB300           | VR200           |
|-----------------------------------|-----------------|-----------------|
| GPU                               | \$100           | \$200           |
| CPU                               | \$50            | \$60            |
| NVSwitch ASIC                     | \$30            | \$30            |
| BlueField DPU                     | \$30            | \$30            |
| ConnectX chip                     | \$30            | \$30            |
| ABF Units per rack                | GB300           | VR200           |
| GPU                               | 72x             | 72x             |
| CPU                               | 36x             | 36x             |
| NVSwitch ASIC                     | 18x             | 36x             |
| BlueField DPU                     | 18x             | 18x             |
| ConnectX chip                     | 36x             | 72x             |
| Total ABF Content per rack        | GB300           | VR200           |
| GPU                               | \$7,200         | \$14,400        |
| CPU                               | \$1,800         | \$2,160         |
| NVSwitch ASIC                     | \$540           | \$1,080         |
| BlueField DPU                     | \$540           | \$540           |
| ConnectX chip                     | \$1,080         | \$2,160         |
| <b>Total ABF Content per rack</b> | <b>\$11,160</b> | <b>\$20,340</b> |

Source: Morgan Stanley Research estimates.



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## Our bottom-up analysis shows cooling component content increasing ~12% for VR200

| Thermal value per Vera Rubin NVL 144 rack   |      |                  |
|---------------------------------------------|------|------------------|
| Compute tray (Bianca)                       | unit | BoM (US\$)       |
| Cold plate module - Compute Board           |      | \$ 400           |
| - # per compute tray                        | 2    | \$ 800           |
| NVQD                                        |      | \$ 20            |
| - # per compute tray                        | 4    | \$ 80            |
| Tray manifold                               |      | \$ 1,000         |
| Cold plate module - Bottom Half             |      | \$ 150           |
| - # per compute tray                        | 3    | \$ 450           |
| NVQD                                        |      | \$ 20            |
| - # per compute tray                        | 16   | \$ 320           |
| Floating mount                              |      | 50               |
| - # per compute tray                        | 2    | \$ 100           |
| Thermal value per compute tray              |      | \$ 2,750         |
| - # of compute tray per rack                | 18   |                  |
| Thermal value for compute tray per rack     |      | \$ 49,500        |
| Switch tray                                 | unit | BoM (US\$)       |
| Cold plate module                           |      | \$ 200           |
| - # per switch tray                         | 2    | \$ 400           |
| NVQD                                        |      | \$ 20            |
| - # per compute tray                        | 18   | \$ 360           |
| Floating mount                              |      | 50               |
| - # per compute tray                        | 2    | \$ 100           |
| Others                                      |      | \$ 60            |
| Thermal value per compute tray              |      | \$ 920           |
| - # of switch tray per rack                 | 9    |                  |
| Thermal value for switch tray per rack      |      | \$ 8,280         |
| <b>total thermal content value per rack</b> |      | <b>\$ 57,780</b> |

| Cooling content (US\$)                 | GB300            | VR200            |
|----------------------------------------|------------------|------------------|
| Cooling (In-tray only)                 | \$50,310         | \$57,780         |
| Rack Manifold                          | \$13,500         | \$13,500         |
| Rack-level other cooling               | \$800            | \$800            |
| <b>Total cooling (ex-Side-car CDU)</b> | <b>\$64,610</b>  | <b>\$72,080</b>  |
| Side-car CDU                           | \$50,000         | \$50,000         |
| <b>Total cooling</b>                   | <b>\$114,610</b> | <b>\$122,080</b> |

Source: Morgan Stanley Research estimates.

Morgan Stanley

## Our bottom-up analysis shows power supply content increasing ~32% for VR200

|                                    | Current           |                   |                   |                   | 2026                       |                    | 2027 |  |
|------------------------------------|-------------------|-------------------|-------------------|-------------------|----------------------------|--------------------|------|--|
| Server power supply design         | Power shelf       |                   |                   |                   | HVDC Standalone power rack |                    |      |  |
| AC-DC conversion                   | 400V AC >> 50V DC | 400V AC >> 50V DC | 400V AC >> 50V DC | 400V AC >> 50V DC | 400V AC >> 800V DC         | 400V AC >> 800V DC |      |  |
| Nvidia AI GPU generation           | GB200             | GB300             | GB300             | Vera Rubin        | Vera Rubin CPX version     | Vera Rubin Ultra   |      |  |
| Nvidia AI server rack architecture | Oberon            | Oberon            | Oberon            | Oberon            | Oberon                     | Kyber              |      |  |
| Power wattage per server rack      | 120kW             | 140kW             | 140kW             | 200kW+            | 380kW+                     | 600kW              |      |  |
| Power wattage per PSU              | 5.5kW             | 8kW               | 12kW              | 18.3kW            | 18.3kW                     | 30kW               |      |  |
| Power value per AI server rack     | US\$36,000 (x)    | US\$57,600        | US\$69,120        | US\$76,000        | US\$398,160                | >10x               |      |  |
| Power value per watt               | US\$0.3           | US\$0.41          | US\$0.49          | US\$0.38          | US\$1.05                   | ...                |      |  |

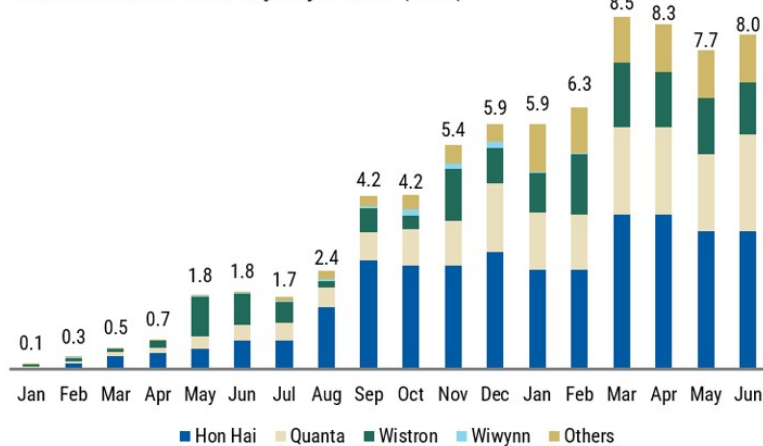
Source: Morgan Stanley Research estimates.

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## We Forecast 2026e GB200/300 Racks To Be 70-80K

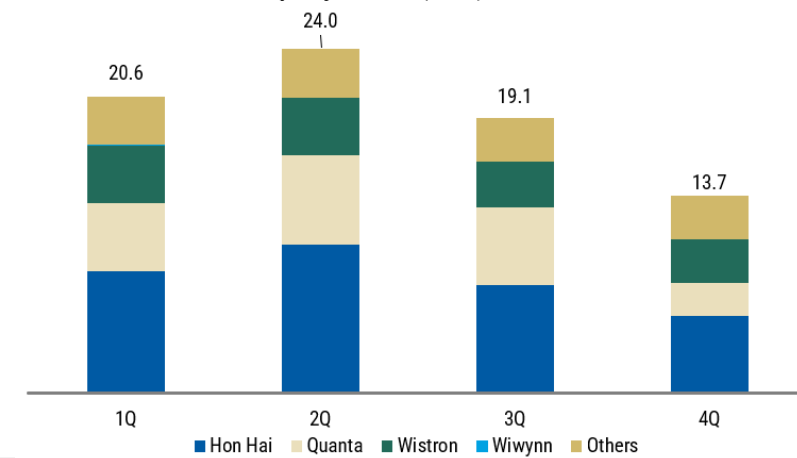
### GB200/300 NVL72-equivalent monthly rack output

GB200/300 NVL72 racks by major ODMs (000s)



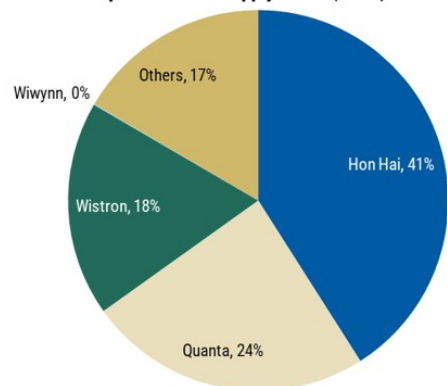
### GB200/300 NVL72-equivalent quarterly rack output (2026e)

GB200/300 NVL72 racks by major ODMs (000s)



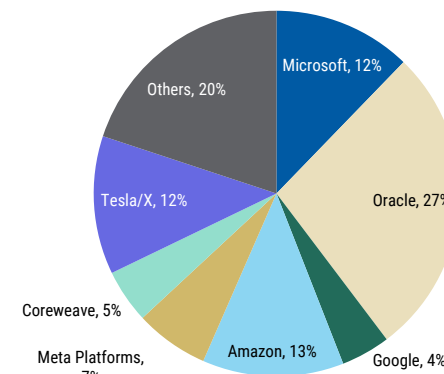
### GB200/300 rack supply share by major ODMs (2026e)

GB200/300 NVL72-equivalent rack supply share (2026)



### GB200/300 rack demand share (2026e)

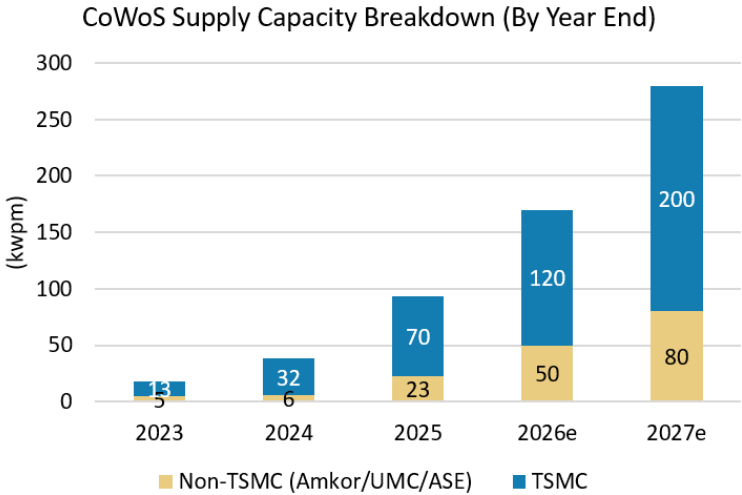
Nvidia GB200/300 server demand share, 2026e



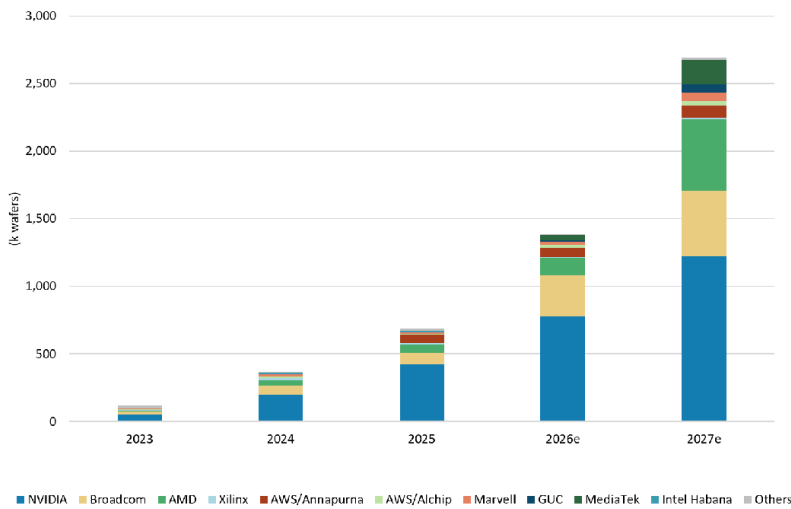
Source: Morgan Stanley Research estimates (e).

# TSMC May Expand CoWoS Capacity to 200Kwpm by 2027 Given Strong AI Demand

CoWoS Supply Capacity Breakdown (By Year End)



Global CoWoS Capacity Demand by Key Customer

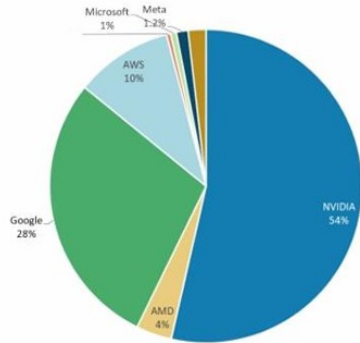


Source: Company data, TrendForce, Morgan Stanley Research estimates (e).

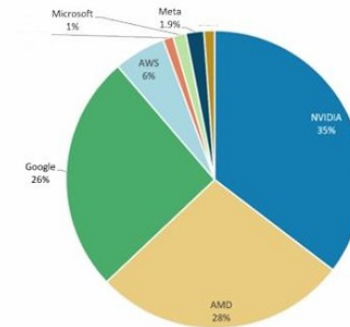
Morgan Stanley

## AI Computing Wafer Consumption Could Rise to US\$46B in CY27

AI semi wafer consumption, by customer, 2026e



AI semi wafer consumption, by customer, 2027e



| AI chip vendor         | Product name                  | CoWoS capacity allocation (k wafers) | Chips per CoWoS wafer | Implied shipments (k) | Compute die size | Geometry | Compute die units | Wafer consumption (k wafers) | Wafer price (US\$) | Wafer revenue TAM (US\$ mn) | Backend revenue (US\$ mn) |
|------------------------|-------------------------------|--------------------------------------|-----------------------|-----------------------|------------------|----------|-------------------|------------------------------|--------------------|-----------------------------|---------------------------|
| <b>AI GPU (2027e)</b>  |                               |                                      |                       |                       |                  |          |                   |                              |                    |                             |                           |
| NVIDIA                 | B300                          | 40                                   | 14                    | 560                   | 850              | 4nm      | 2                 | 44                           | 23,042             | 1,024                       | 845                       |
|                        | Vera CPU                      | 250                                  | 23                    | 5,750                 | 850              | 3nm      | 1                 | 228                          | 27,300             | 6,229                       | 5,283                     |
|                        | Rubin R200                    | 740                                  | 8                     | 5,920                 | 850              | 3nm      | 2                 | 470                          | 27,300             | 12,827                      | 15,637                    |
|                        | Rubin Ultra                   | 130                                  | 8                     | 1,040                 | 850              | 3nm      | 2                 | 58                           | 27,300             | 1,588                       | 2,210                     |
| AMD                    | MI350 series                  | 24                                   | 12                    | 288                   | 110              | 3nm      | 8                 | 8                            | 27,300             | 229                         | 667                       |
|                        | MI455                         | 157                                  | 7                     | 1,099                 | 850              | 2nm      | 2                 | 13                           | 30,000             | 400                         | 4,360                     |
|                        | MI450 (for Meta)              | 35                                   | 14                    | 490                   | 850              | 2nm      | 1                 | 3                            | 30,000             | 89                          | 972                       |
|                        | MI500 (Arcadia)               | 24                                   | 4                     | 96                    |                  | 2nm      |                   |                              | 30,000             |                             |                           |
|                        | Venice CPU                    | 270                                  | 21                    | 5,670                 |                  | 2nm      |                   |                              |                    |                             |                           |
| <b>AI ASIC (2027e)</b> |                               |                                      |                       |                       |                  |          |                   |                              |                    |                             |                           |
| Google                 | TPU v7p (Ironwood; AVGO)      | 13                                   | 16                    | 208                   | 700              | 3nm      | 2                 | 14                           | 27,300             | 371                         | 298                       |
|                        | TPU v8i (Sunfish; AVGO)       | 330                                  | 12                    | 3,960                 | 800              | 3nm      | 2                 | 296                          | 27,300             | 8,075                       | 7,571                     |
|                        | TPU v8t (Zebrafish; MediaTek) | 180                                  | 20                    | 3,600                 | 800              | 3nm      | 2                 | 269                          | 27,300             | 7,341                       | 4,130                     |
|                        | TPU v9 (Humafish; MediaTek)   |                                      |                       | 400                   | 850              | 2nm      | 2                 | 32                           | 27,300             | 867                         |                           |
| AWS                    | Trainium 3                    | 140                                  | 17                    | 2,380                 | 700              | 3nm      | 2                 | 127                          | 27,300             | 3,465                       | 2,874                     |
| GUC's customers        |                               | 60                                   | 20                    | 1,200                 | 645              | 4nm      | 1                 | 29                           | 23,042             | 674                         | 30                        |
| Microsoft              | Maia 300                      | 50                                   | 11                    | 550                   | 850              | 2nm      | 1                 | 29.1                         | 30,000             | 873                         | 1,087                     |
| Meta                   | MTIA 3 (Iris)                 | 55                                   | 10                    | 550                   | 850              | 3nm      | 2                 | 44                           | 27,300             | 1,192                       | 1                         |
| Others                 |                               | 18                                   |                       |                       |                  |          |                   |                              |                    |                             |                           |
| <b>Total</b>           |                               | <b>2,664</b>                         |                       |                       |                  |          |                   | <b>1,700</b>                 |                    | <b>46,208</b>               | <b>45,965</b>             |

Source: Morgan Stanley Research estimates.

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## More ASIC Projects Are Coming, According to Each CSP Plan

| k Units                                 | 2023       | 2024         | 2025e        | 2026e        | 2027e        | 2028e           |
|-----------------------------------------|------------|--------------|--------------|--------------|--------------|-----------------|
| v5                                      | 500        | 2,400        | 250          |              |              |                 |
| v6 (Trillium)                           |            |              | 1,000        |              |              |                 |
| v7 (Ironwood, by Broadcom)              |            |              | 500          | 2,300        | 500          |                 |
| v8i (Sunfish; 3nm, by Broadcom)         |            |              |              | 900          | 3,000        | 2,500           |
| v8t (Zebrafish; 3nm, by MediaTek)       |            |              |              | 500          | 3,000        | 1,000           |
| v9 (Humufish; 2nm, by MediaTek)         |            |              |              |              | 150          | 3,000           |
| v9a (Merope; 2nm, by US design service) |            |              |              |              |              | unknown         |
| v10 (Icefish; 1.4nm, by MediaTek)       |            |              |              |              |              | unknown         |
| <b>Total</b>                            | <b>500</b> | <b>2,400</b> | <b>1,750</b> | <b>3,700</b> | <b>6,650</b> | <b>&gt;6500</b> |

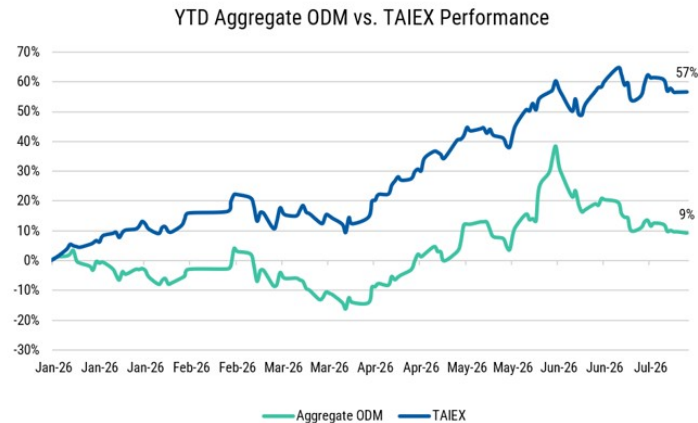
Source: Morgan Stanley Research estimates.



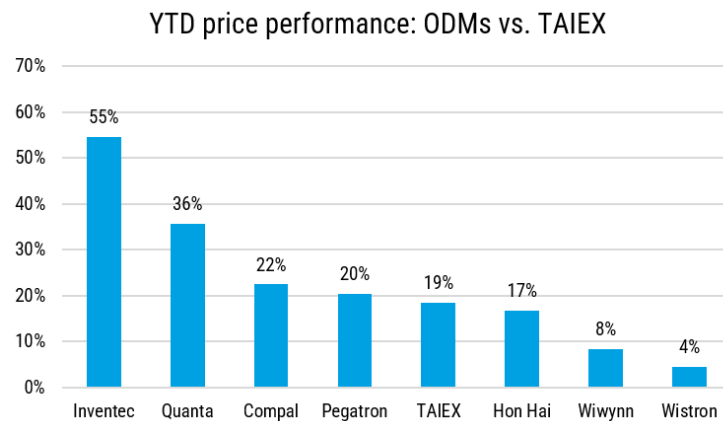
Morgan Stanley

## ODMs Have Underperformed YTD

### ODM Cohort Has Underperformed TAIEX by more than 40pts YTD

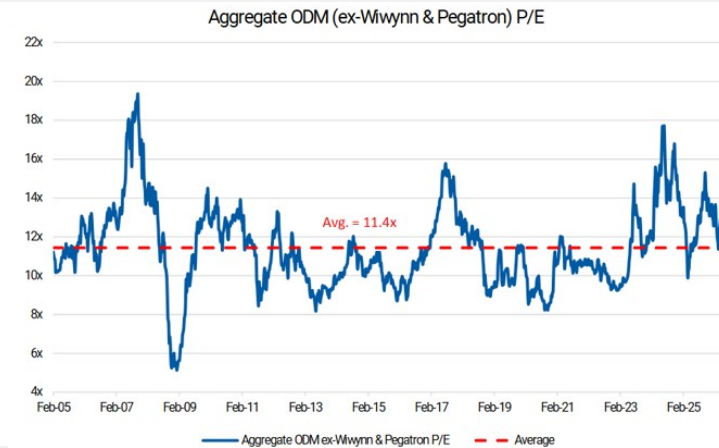


### YTD Price Performance – Individual ODMs vs TAIEX



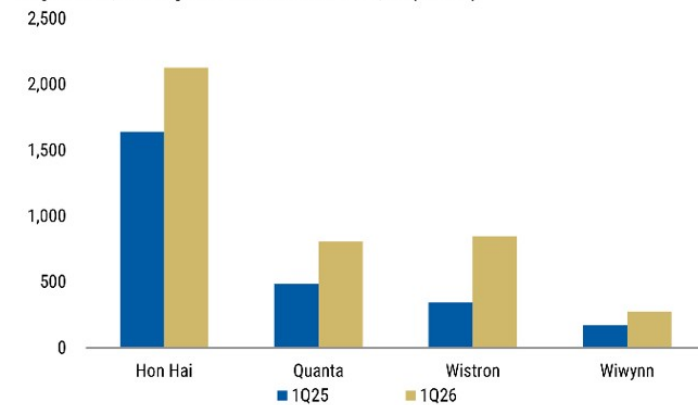
Source: Morgan Stanley Research.

### ODM Cohort Is Trading At Avg Historical P/E

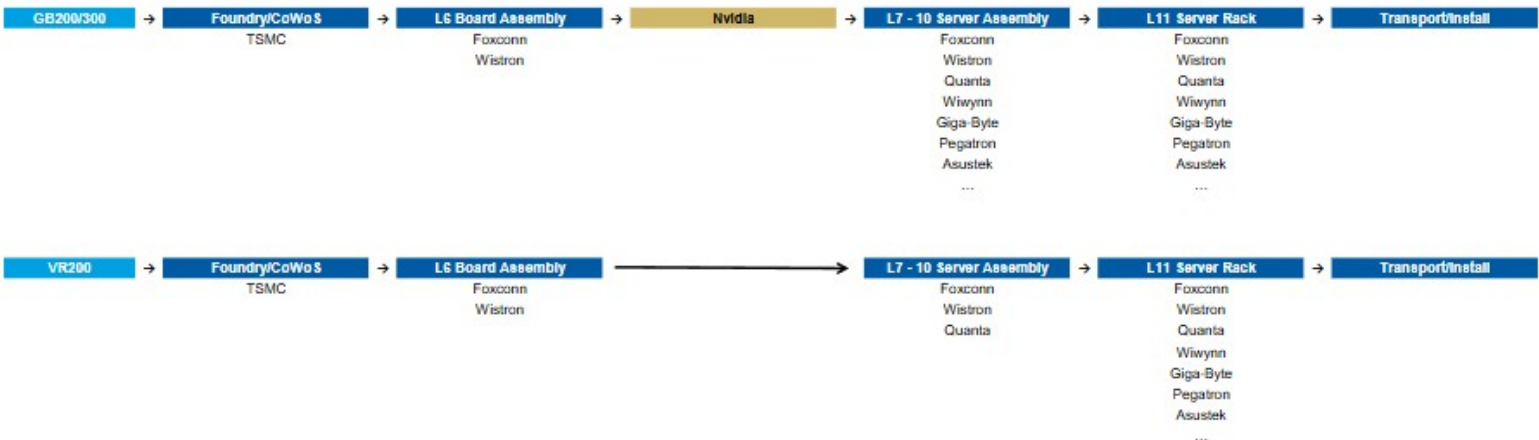


### ...While Revenue Continues to Grow y/y

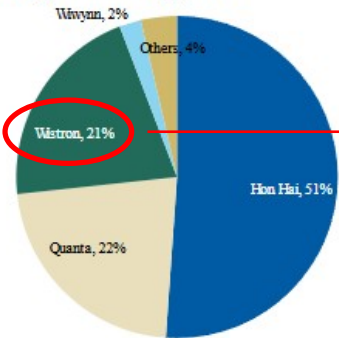
#### Key ODM Quarterly Revenue in C1Q25-1Q26 (NT\$ B)



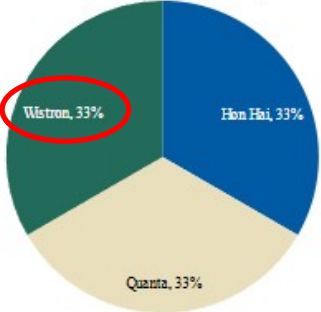
Wistron Is a Share Winner from VR200 Onward as Well as MI400-series



GB200/300 NVL72-equivalent rack supply share



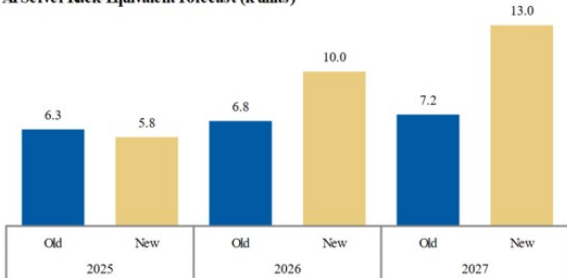
Morgan Stanley Estimate of Compute Tray Share (VR200)



Wistron Sensitivity Analysis for GB300

| EPS contribution  |     | Rack shipment (000s) |      |      |      |      |      |
|-------------------|-----|----------------------|------|------|------|------|------|
| Rack ASP (US\$ M) |     | 1                    | 2    | 3    | 4    | 5    | 6    |
|                   | 2.6 | 0.36                 | 0.73 | 1.09 | 1.46 | 1.82 | 2.18 |
|                   | 2.8 | 0.39                 | 0.78 | 1.18 | 1.57 | 1.96 | 2.35 |
|                   | 3.0 | 0.42                 | 0.84 | 1.26 | 1.68 | 2.10 | 2.52 |
|                   | 3.2 | 0.45                 | 0.90 | 1.34 | 1.79 | 2.24 | 2.69 |
|                   | 3.4 | 0.48                 | 0.95 | 1.43 | 1.90 | 2.38 | 2.85 |
|                   | 3.6 | 0.50                 | 1.01 | 1.51 | 2.01 | 2.52 | 3.02 |

AI Server Rack-Equivalent Forecast (k units)

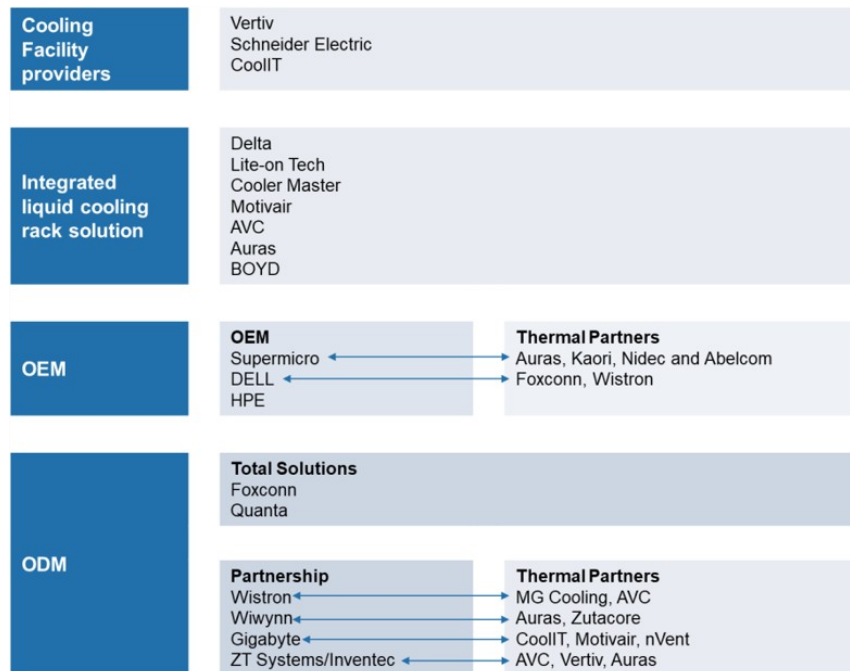


Source: NVIDIA, Morgan Stanley Research.

Morgan Stanley

## Suppliers with Integration Capability and Proven Shipment Record Stand Out

### Liquid cooling solution supply dynamics for GB200 server rack system



### GB200 / ASIC server rack ODM partners

| GB200/300      | ODM Partner(s)                |
|----------------|-------------------------------|
| Microsoft      | Foxconn/Quanta                |
| Oracle         | Foxconn/Wistron/Wiwynn/Quanta |
| Google         | Quanta/Foxconn/Inventec       |
| Amazon         | Quanta/Foxconn                |
| Meta Platforms | Quanta/Foxconn                |
| Dell           | Wistron                       |
| HPE            | Foxconn/Wistron               |
| Others         | SMCI / etc                    |

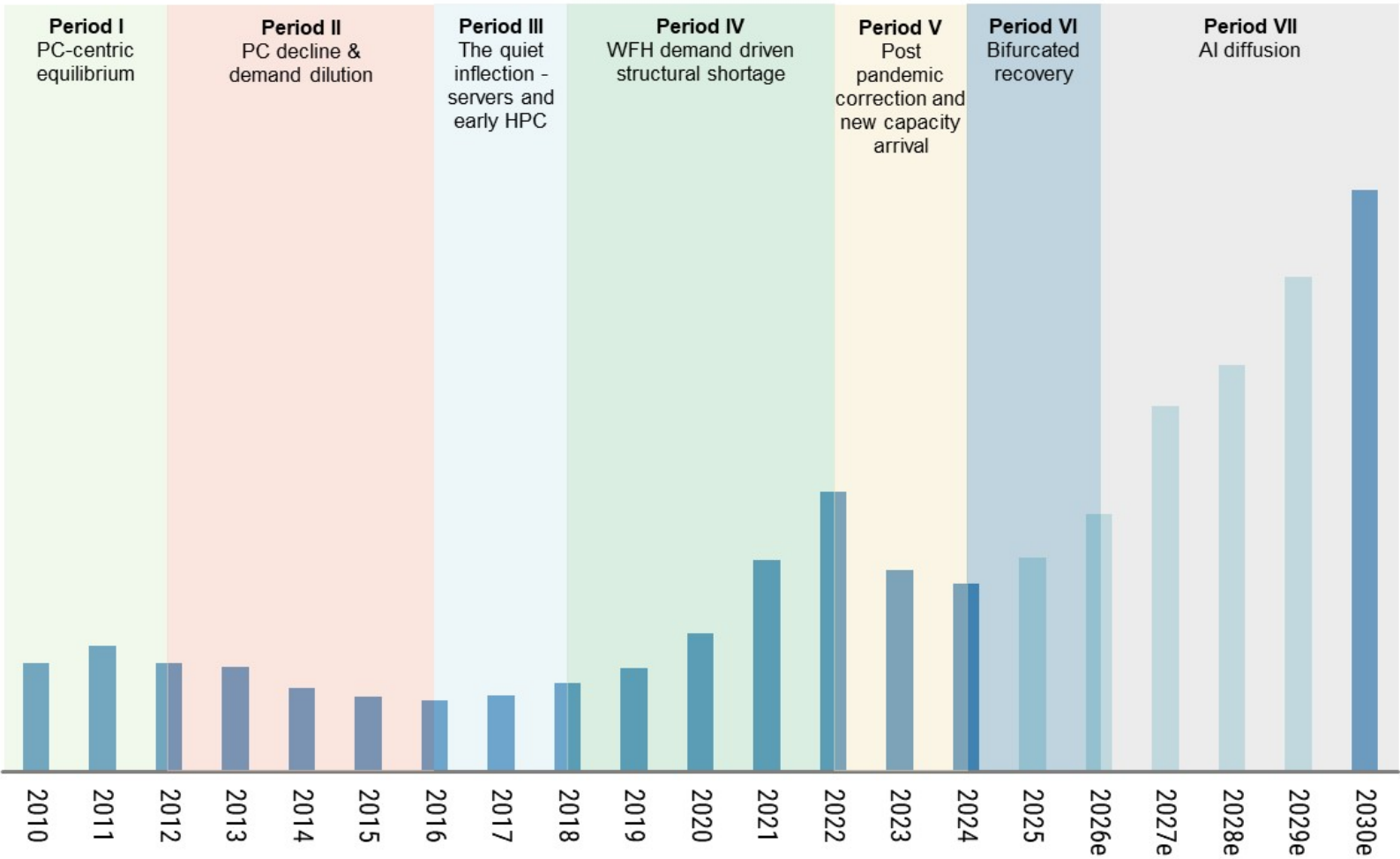
  

| ASIC           | ODM Partner(s)             |
|----------------|----------------------------|
| Amazon         | Wiwynn/Jabil               |
| Google         | Celestica/Foxconn/Inventec |
| Meta Platforms | Quanta/Celestica/Wiwynn    |

Source: Morgan Stanley Research.

Morgan Stanley

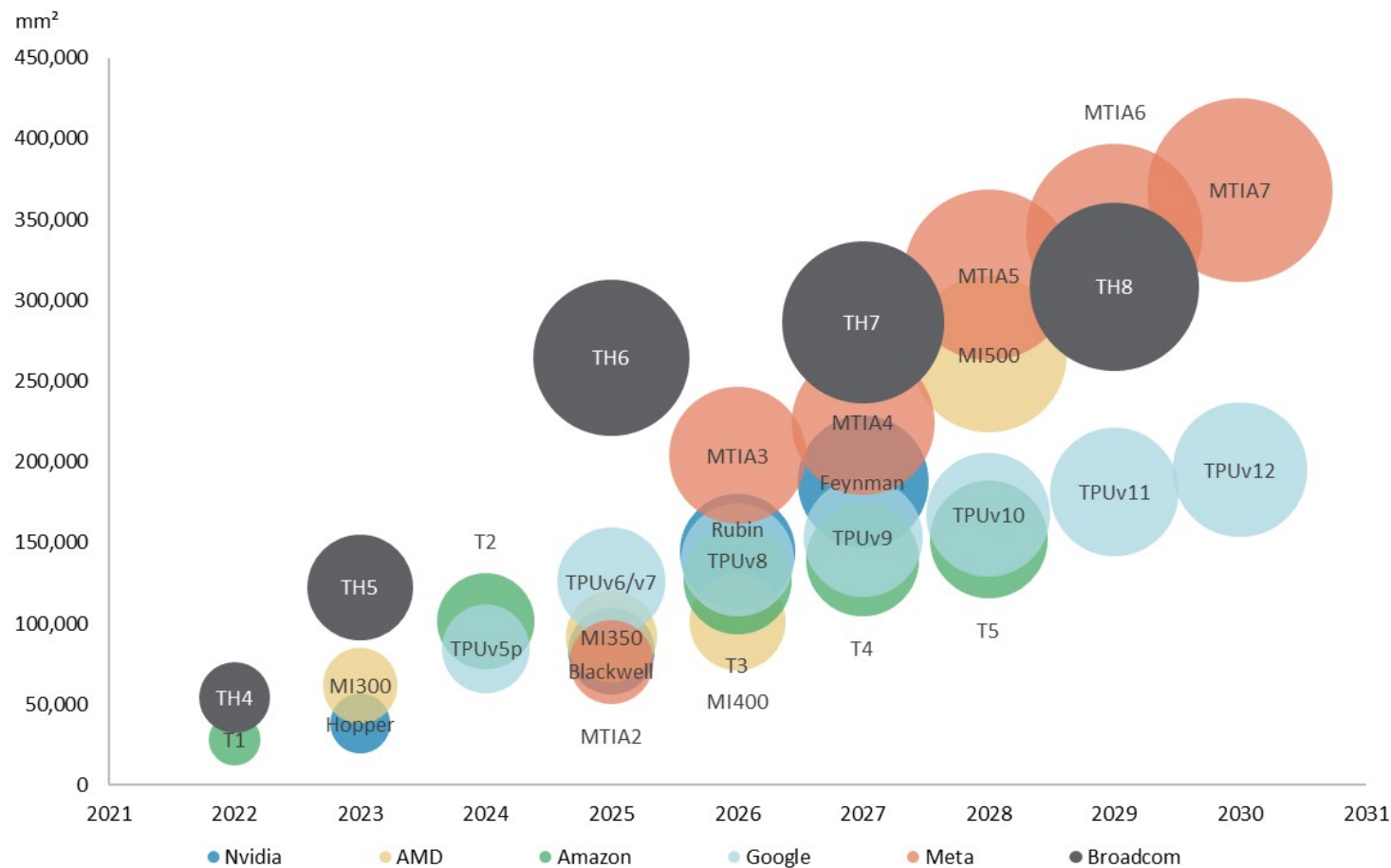
ABF Substrate: We Are Entering the AI Diffusion Era



Source: Company data, Morgan Stanley Research estimates.

Morgan Stanley

# ABF Substrate Body Size and Layer Count Continue to Increase

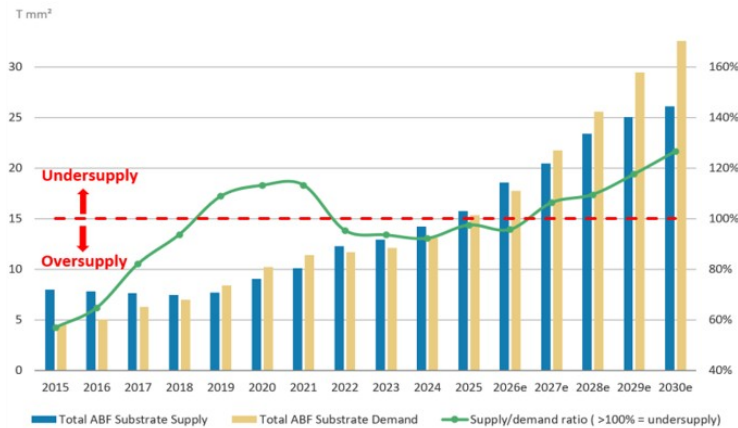


Source: Company data, Morgan Stanley Research estimates.

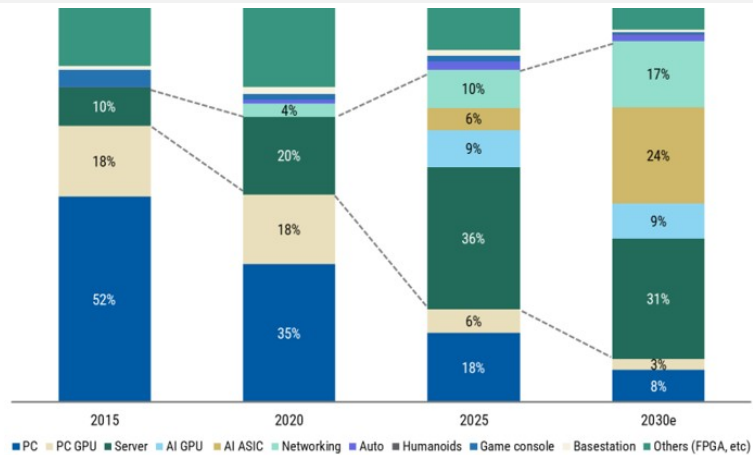
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## ABF Substrates: Undersupply from 2027 Onward

### Expect ABF substrate undersupply from 2027 onwards

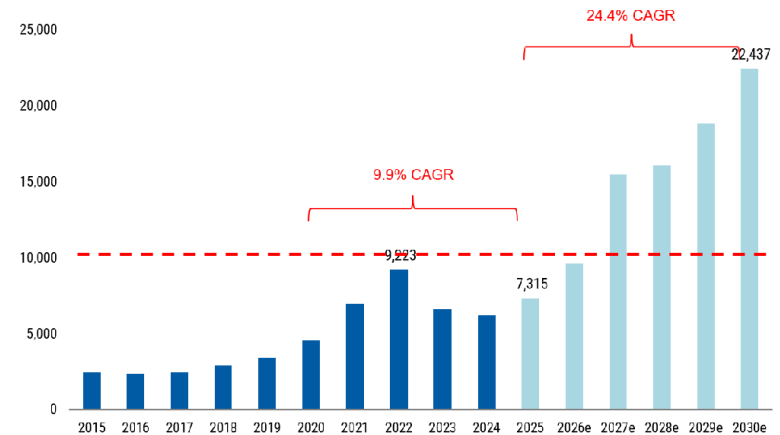


### AI-related applications expected to rise to 75%+ by 2030



Source: Prismark, Morgan Stanley Research. e = Morgan Stanley Research estimates.

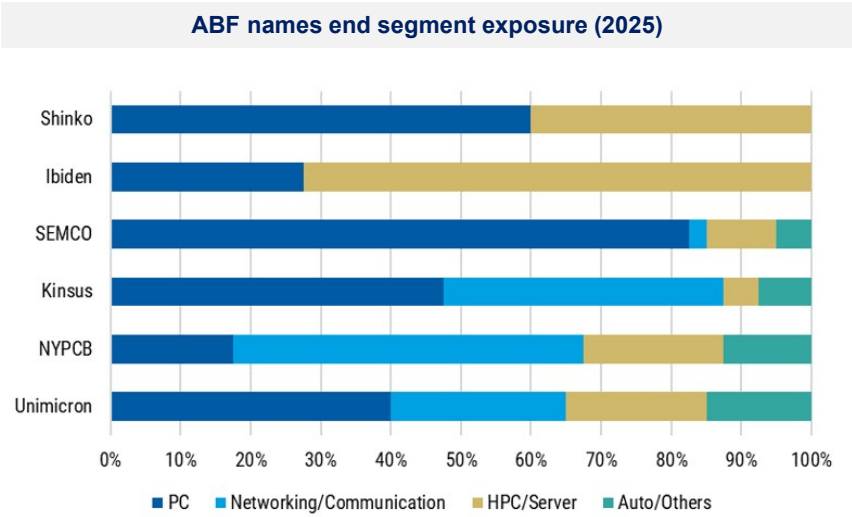
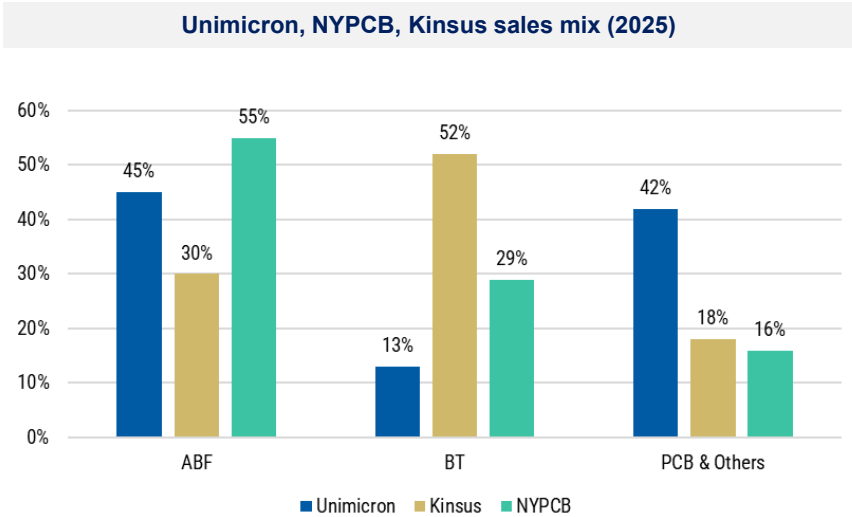
### We are in early innings of an AI-driven upcycle



### Expect T glass to remain undersupplied until 2028



# ABF Substrates: Sales Mix by Product and End Segment



Source: Company data, Morgan Stanley Research estimates.



## NVIDIA GPU ABF/CCL/PCB Roadmap

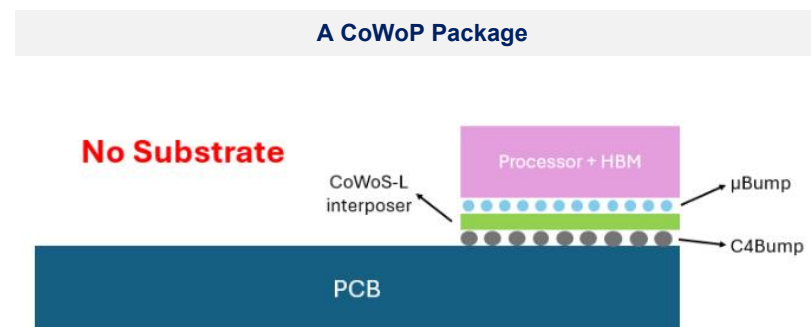
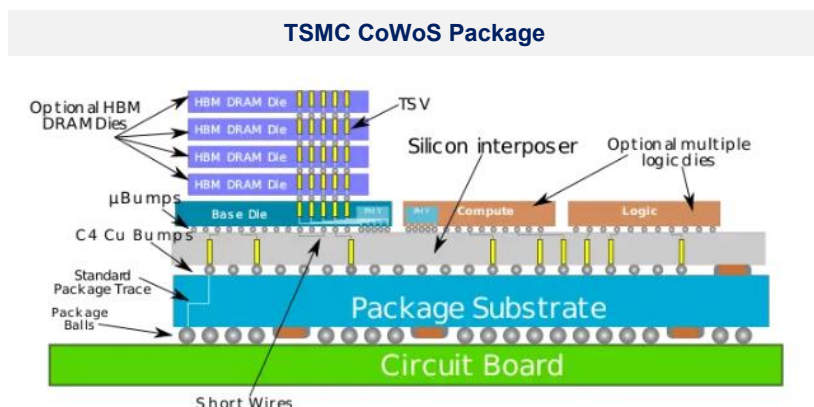
| NVIDIA                              | A100                       | H100                                         | B200                                            |
|-------------------------------------|----------------------------|----------------------------------------------|-------------------------------------------------|
| <b>ABF Substrate</b>                |                            |                                              |                                                 |
| Substrate size                      | 55mm x 55mm                | 55mm x 58mm                                  | 73mm x 81mm                                     |
| Layer count                         | 12L                        | 12L                                          | 14L                                             |
| Supplier                            | Ibiden                     | Ibiden                                       | Ibiden/Unimicron                                |
| <b>OCP Accelerator Module (OAM)</b> |                            |                                              |                                                 |
| CCL Spec.                           | 0                          | Very low -loss (M6),<br>Ultra low -loss (M7) | Extreme low -loss (M8) +<br>Low -loss (M4)      |
| CCL supplier                        | 0                          | EMC                                          | Doosan                                          |
| PCB Size                            | 140mm x 78mm               | 152mm x 80.5mm                               | 170mm x 86.5mm                                  |
| PCB Spec.                           | 12L                        | 18L HDI                                      | 20L HDI                                         |
| PCB Supplier                        | 0                          | UMTC, ISU, TTM, WUS                          | UMTC, VGT, ISU, TTM                             |
| <b>Universal Baseboard (UBB)</b>    |                            |                                              |                                                 |
| CCL Spec.                           | 0                          | Ultra low -loss (M7)                         | Extreme low -loss (M8) +<br>Low -loss (M4)      |
| CCL supplier                        | EMC                        | EMC, TUC                                     | Doosan, TUC                                     |
| PCB Spec.                           | 0                          | 24L PTH                                      | 18L PTH                                         |
| PCB Supplier                        | 0                          | WUS, TTM, ISU                                | WUS, TTM, ISU, UMTC                             |
| <b>Switch board</b>                 |                            |                                              |                                                 |
| CCL Spec.                           | Ultra low -loss (M7)       | Very low -loss (M6),<br>Ultra low -loss (M7) | Ultra low -loss (M7),<br>Extreme low -loss (M8) |
| CCL supplier                        | 0                          | EMC, TUC                                     | EMC, TUC, Doosan                                |
| PCB Spec.                           | 0                          | 26-28L                                       | 28-30L                                          |
| PCB Supplier                        | 0                          | WUS, TTM, GCE, ISU                           | WUS, TTM, GCE, ISU                              |
| <b>Motherboard</b>                  |                            |                                              |                                                 |
| CCL Spec.                           | Very low -loss (M6)        | Very low -loss (M6)                          | Very low -loss (M6)                             |
| CCL supplier                        | EMC, TUC, ITEQ, Panasonic  | EMC, TUC,<br>ITEQ, Panasonic                 | EMC, TUC,<br>ITEQ, Panasonic                    |
| PCB Spec.                           | 16-18L                     | 16-18L                                       | 16-20L                                          |
| PCB Supplier                        | GCE, WUS, TTM, ISU, Tripod | GCE, WUS, TTM,<br>ISU, Tripod                | GCE, WUS, TTM,<br>ISU, Tripod                   |

Source: Morgan Stanley Research.

| NVIDIA                                  | GB200 NVL72                                | GB300 NVL72                                | VR200                                      |
|-----------------------------------------|--------------------------------------------|--------------------------------------------|--------------------------------------------|
| <b>ABF Substrate</b>                    |                                            |                                            |                                            |
| Substrate size                          | 73mm x 81mm                                | 73mm x 81mm                                | 83mm x 97mm                                |
| Layer count                             | 14L                                        | 14L                                        | 18L                                        |
| Supplier                                | Ibiden/Unimicron                           | Ibiden/Unimicron                           | Ibiden/Unimicron                           |
| <b>OCP Accelerator Module (OAM)</b>     |                                            |                                            |                                            |
| CCL Spec.                               | Extreme low -loss (M8) +<br>Low -loss (M4) | Extreme low -loss (M8) +<br>Low -loss (M4) | Extreme low -loss (M8) +<br>Low -loss (M4) |
| CCL supplier                            | Doosan                                     | Doosan                                     | Doosan                                     |
| PCB Size                                | 305.5mm x 222mm                            | 305.5mm x 222mm                            | 305.5mm x 222mm                            |
| PCB Spec.                               | 22L HDI                                    | 22L HDI                                    | 26L HDI                                    |
| PCB Supplier                            | VGT, UMTC                                  | VGT, UMTC                                  | VGT, UMTC,<br>ZDT, Dynamic                 |
| <b>Switch board</b>                     |                                            |                                            |                                            |
| CCL Spec.                               | Extreme low -loss (M8)                     | Extreme low -loss (M8)                     | Extreme low -loss (M8)                     |
| CCL supplier                            | EMC                                        | EMC, Shengyi                               | EMC, Shengyi                               |
| PCB Spec.                               | 24L PTH                                    | 24L PTH                                    | 32L PTH                                    |
| PCB Supplier                            | WUS, TTM, VGT                              | VGT, TTM, Kinw on                          | VGT, WUS, TTM                              |
| <b>Midplane PCB</b>                     |                                            |                                            |                                            |
| CCL Spec.                               | NA                                         | NA                                         | Beyond extreme low -loss<br>(M9)           |
| CCL supplier                            | NA                                         | NA                                         | EMC                                        |
| PCB Spec.                               | NA                                         | NA                                         | 44L PTH                                    |
| PCB Supplier                            | NA                                         | NA                                         | VGT, WUS                                   |
| <b>Orchid Module / BlueField Module</b> |                                            |                                            |                                            |
| CCL Spec.                               | NA                                         | NA                                         | Extreme low -loss (M8)                     |
| CCL supplier                            | NA                                         | NA                                         | EMC                                        |
| PCB Spec.                               | NA                                         | NA                                         | 20L PTH                                    |
| PCB Supplier                            | NA                                         | NA                                         | WUS, VGT, TTM, UMTC                        |

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## AI GPUs to adopt CoWoP instead of CoWoS, reducing reliance on ABF substrates?



**What is CoWoP?** CoWoP stands for Chip on Wafer on PCB, in which a PCB (printed circuit board) replaces the ABF substrate in a CoWoS (Chip on Wafer on Substrate) IC package. Currently, all Nvidia's data center GPUs adopt TSMC's CoWoS technology, and TSMC's CoWoS yield rate is close to 100%.

**Why use CoWoP if CoWoS yield rate is so high?** By adopting CoWoP, the aim is to resolve substrate warpage, increase NVLink reach on the PCB without substrate in between chip and PCB, achieve greater cooling efficiency without a package lid, and remove capacity bottleneck in certain package materials.

**We don't think Rubin Ultra will use CoWoP:** In order to adopt CoWoP, the L/S (line/space) of the PCB needs to shrink to  $<10/10\mu\text{m}$ , because that is where L/S for ABF substrates currently is. Average high-density interconnect (HDI) PCB is  $40/50\mu\text{m}$  today. SLP (substrate-like PCB), which is used in iPhone mainboards today, is  $20/35\mu\text{m}$  – hence the name SLP, because its L/S is closer to that of substrates. However, to get the PCB L/S from  $20/35\mu\text{m}$  to below  $10/10\mu\text{m}$  would be difficult, which is one reason why we don't think it will happen for Rubin Ultra. This is in line with Shoji Sato's checks as well, with Ibmiden saying that Rubin Ultra's ABF substrate is much larger than Rubin's, with more layers. We believe shifting to CoWoP from CoWoS would entail significant yield risks and reshuffling of the related supply chain, which does not look rational to us for products targeted to enter mass production in one year's time.

Source: Company data, Morgan Stanley Research.

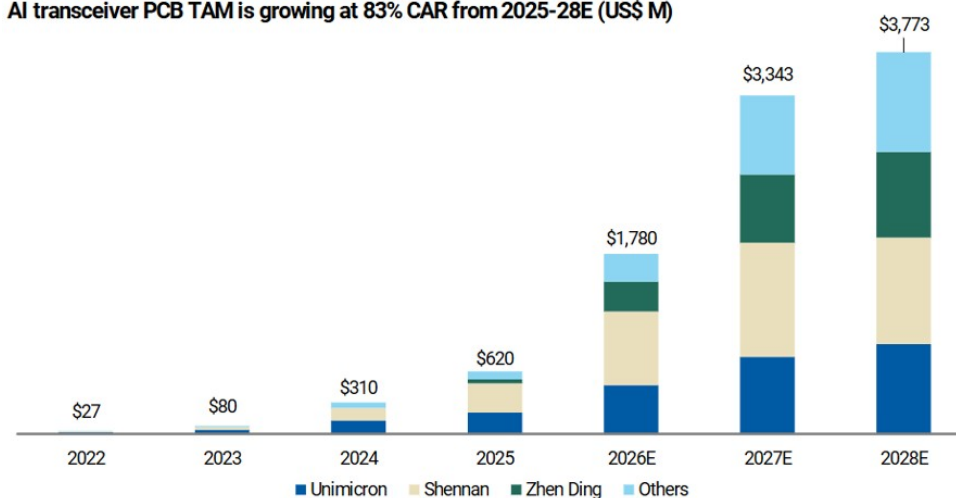
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## Optical transceiver – PCB beneficiaries of AI interconnect build-out

| <i>(mn units)</i>  |            |            |            |
|--------------------|------------|------------|------------|
| New Estimates      | 2026E      | 2027E      | 2028E      |
| 800G               | 44         | 63         | 64         |
| 1.6T               | 29         | 79         | 87         |
| 3.2T               | 0          | 0          | 8          |
| <b>Total</b>       | <b>73</b>  | <b>141</b> | <b>158</b> |
| Previous Estimates | 2026E      | 2027E      | 2028E      |
| 800G               | 34         | 48         | 54         |
| 1.6T               | 19         | 24         | 27         |
| 3.2T               | 0          | 0          | 5          |
| <b>Total</b>       | <b>53</b>  | <b>71</b>  | <b>85</b>  |
| Pct. of change     | 2026E      | 2027E      | 2028E      |
| 800G               | 29%        | 31%        | 19%        |
| 1.6T               | 52%        | 233%       | 226%       |
| 3.2T               | --         | --         | 60%        |
| <b>Total</b>       | <b>38%</b> | <b>98%</b> | <b>86%</b> |

|            | 400G                                                 | 800G                                                | 1.6T                                                |
|------------|------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|
| Layers     | 10-12L<br>2-3 press HDI                              | 12-14L, 4-8L mSAP<br>or HDI                         | 14-16L<br>6-10L mSAP                                |
| Process    | HDI                                                  | HDI or mSAP                                         | mSAP                                                |
| CCL Spec   | M6                                                   | M7(LDK1) / M7+ (LDK2)                               | M7+ / M8 (LDK2)                                     |
| ASP (US\$) | \$5-15                                               | \$15-25                                             | \$20-30                                             |
| GM Est.    | 20-30%                                               | 30-40%                                              | 40-50%+                                             |
| Suppliers  | Unimicron, Shennan, WUS, other<br>smaller suppliers. | Unimicron, Shennan, Zhen Ding,<br>WUS, Compeq, etc. | Unimicron, Shennan, Zhen Ding,<br>Compeq, WUS, etc. |

AI transceiver PCB TAM is growing at 83% CAR from 2025-28E (US\$ M)



Source: Company data, Morgan Stanley Research estimates.

ABF Substrates: M-series Chips



| Chip        | M1                 | M1 Pro               | M1 Max               | M1 Ultra             | M2                 | M2 Ultra             | M4                 |
|-------------|--------------------|----------------------|----------------------|----------------------|--------------------|----------------------|--------------------|
| Dimensions  | 24 x 27mm          | 57 x 25mm            | 57 x 36mm            | 65 x 72mm            | 24 x 27mm          | 70.6 x 70mm          | 24 x 27.7mm        |
| Size        | 648mm <sup>2</sup> | 1,425mm <sup>2</sup> | 2,052mm <sup>2</sup> | 4,680mm <sup>2</sup> | 648mm <sup>2</sup> | 4,942mm <sup>2</sup> | 665mm <sup>2</sup> |
| Layer count | 8L                 | 10L                  | 10L                  | 18L                  | 8L                 | 18L                  | 8L                 |

Source: Apple, Morgan Stanley Research

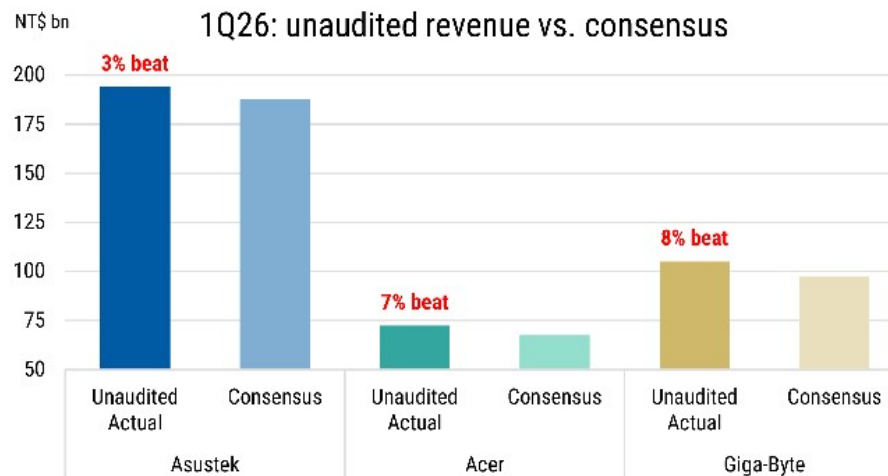
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## What's the Latest on PCs?

### Recent updates

- "Demand is outstripping supply" in the first half of the year.
- PC prices are increasing, and they are increasing faster than costs are rising. OEMs are optimizing their product portfolios to sell less low-end PCs and more high-end PCs due to component constraints.
- We believe this is going to drive the OEMs to see stronger profits and margins in the near term.
- However, we maintain our cautious view and think the margin and demand impact will start to emerge in 2H26.

#### All Taiwan OEMs' Revenue Came In Better-than-Expected in 1Q



Source: Company data, e = Morgan Stanley Research estimates.

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## We keep our 3Q26 NB builds largely unchanged at 28.9M units (-5% q/q, -15% y/y)

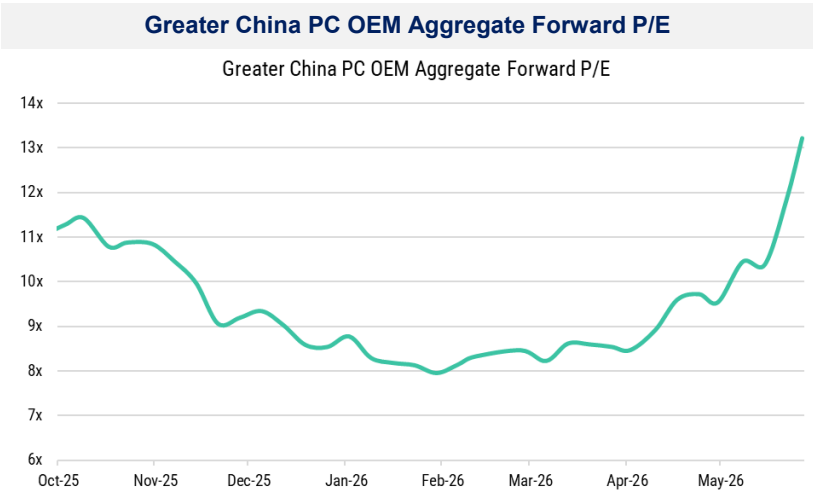
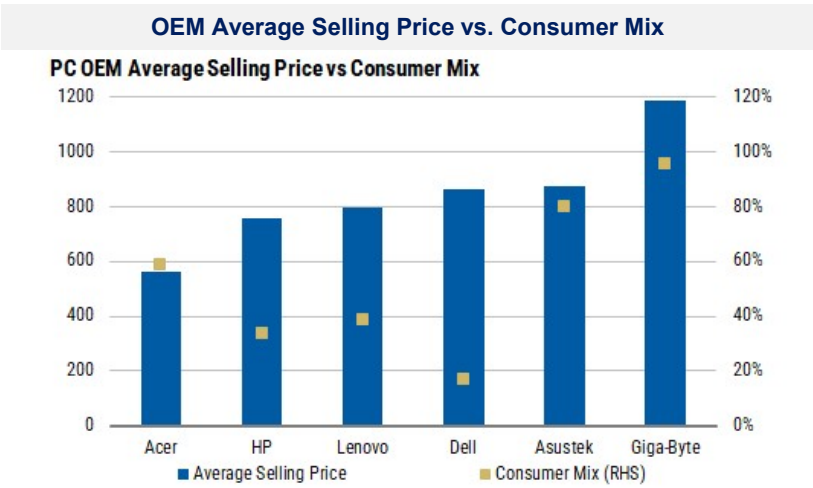
| Shipment (unit: k) | Jan-25 | Feb-25 | Mar-25 | Apr-25 | May-25 | Jun-25 | Jul-25 | Aug-25 | Sep-25 | Oct-25E | Nov-25 | Dec-25 | Jan-26 | Feb-26 | Mar-26 | Apr-26 | May-26 | Jun-26 | Jun-26E | Diff. | Jul-26E |
|--------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|---------|--------|--------|--------|--------|--------|--------|--------|--------|---------|-------|---------|
| Quanta             | 2,800  | 3,100  | 4,900  | 3,300  | 3,800  | 5,000  | 4,000  | 4,100  | 4,600  | 3,500   | 3,500  | 3,900  | 2,700  | 2,000  | 5,300  | 3,500  | 3,500  | 4,500  | 3,500   | 29%   | 3,200   |
| Wistron            | 1,500  | 1,500  | 1,900  | 1,700  | 1,800  | 2,400  | 2,000  | 2,200  | 2,200  | 2,100   | 2,200  | 2,600  | 1,700  | 1,600  | 2,800  | 1,800  | 1,700  | 2,000  | 2,200   | -9%   | 1,500   |
| Others             | 1,500  | 1,600  | 1,900  | 1,700  | 1,800  | 2,100  | 1,700  | 1,800  | 1,900  | 1,600   | 1,700  | 2,000  | 1,700  | 1,500  | 2,200  | 1,500  | 1,500  | 2,400  | 2,200   | 9%    | 1,500   |
| Compal             | 2,000  | 2,100  | 2,900  | 2,200  | 2,400  | 2,500  | 2,300  | 2,300  | 2,500  | 2,100   | 2,300  | 2,400  | 1,600  | 1,500  | 2,800  | 1,800  | 2,000  | 2,400  | 2,200   | 9%    | 1,800   |
| Pegatron           | 610    | 575    | 700    | 610    | 775    | 990    | 750    | 825    | 825    | 775     | 710    | 925    | 700    | 425    | 675    | 525    | 575    | 650    | 750     | -13%  | 550     |
| Total              | 8,410  | 8,875  | 12,300 | 9,510  | 10,575 | 12,990 | 10,750 | 11,225 | 12,025 | 10,075  | 10,410 | 11,825 | 8,400  | 7,025  | 13,775 | 9,125  | 9,275  | 11,950 | 10,850  | 10%   | 8,550   |
| <b>MoM %</b>       |        |        |        |        |        |        |        |        |        |         |        |        |        |        |        |        |        |        |         |       |         |
| Quanta             | -35%   | 11%    | 58%    | -33%   | 15%    | 32%    | -20%   | 2%     | 12%    | -24%    | 0%     | 11%    | -31%   | -26%   | 165%   | -34%   | 0%     | 29%    | 0%      |       | -29%    |
| Wistron            | -17%   | 0%     | 27%    | -11%   | 6%     | 33%    | -17%   | 10%    | 0%     | -5%     | 5%     | 18%    | -35%   | -6%    | 75%    | -36%   | -6%    | 18%    | 29%     |       | -25%    |
| Others             | -25%   | 7%     | 19%    | -11%   | 6%     | 17%    | -19%   | 6%     | 6%     | -16%    | 6%     | 18%    | -15%   | -12%   | 47%    | -32%   | 0%     | 60%    | 47%     |       | -38%    |
| Compal             | -13%   | 5%     | 38%    | -24%   | 9%     | 4%     | -8%    | 0%     | 9%     | -16%    | 10%    | 4%     | -33%   | -6%    | 87%    | -36%   | 11%    | 20%    | 10%     |       | -25%    |
| Pegatron           | -19%   | -6%    | 22%    | -13%   | 27%    | 28%    | -24%   | 10%    | 0%     | -6%     | -8%    | 30%    | -24%   | -39%   | 59%    | -22%   | 10%    | 13%    | 30%     |       | -15%    |
| Total              | -25%   | 6%     | 39%    | -23%   | 11%    | 23%    | -17%   | 4%     | 7%     | -16%    | 3%     | 14%    | -29%   | -16%   | 96%    | -34%   | 2%     | 29%    | 17%     |       | -28%    |
| <b>YoY%</b>        |        |        |        |        |        |        |        |        |        |         |        |        |        |        |        |        |        |        |         |       |         |
| Quanta             | -3%    | 7%     | 4%     | 3%     | -5%    | 11%    | 18%    | -5%    | -6%    | 9%      | -3%    | -9%    | -4%    | -35%   | 8%     | 6%     | -8%    | -10%   | -30%    |       | -20%    |
| Wistron            | 15%    | 7%     | 0%     | 6%     | 6%     | 33%    | 25%    | 29%    | 16%    | 17%     | 29%    | 44%    | 13%    | 7%     | 47%    | 6%     | -6%    | -17%   | -8%     |       | -25%    |
| Others             | 0%     | 23%    | 12%    | 13%    | 6%     | 24%    | 0%     | 6%     | 6%     | -6%     | 0%     | 0%     | 13%    | -6%    | 16%    | -12%   | -17%   | 14%    | 5%      |       | -12%    |
| Compal             | -9%    | -9%    | -3%    | -21%   | -17%   | -17%   | -8%    | -21%   | -14%   | -25%    | -15%   | 4%     | -20%   | -29%   | -3%    | -18%   | -17%   | -4%    | -12%    |       | -22%    |
| Pegatron           | 16%    | 53%    | 8%     | 16%    | 6%     | 28%    | -3%    | -6%    | 0%     | 35%     | 14%    | 23%    | 15%    | -26%   | -4%    | -14%   | -26%   | -34%   | -24%    |       | -27%    |
| Total              | 0%     | 7%     | 3%     | -1%    | -4%    | 10%    | 8%     | -2%    | -2%    | 0%      | 1%     | 6%     | 0%     | -21%   | 12%    | -4%    | -12%   | -8%    | -16%    |       | -20%    |

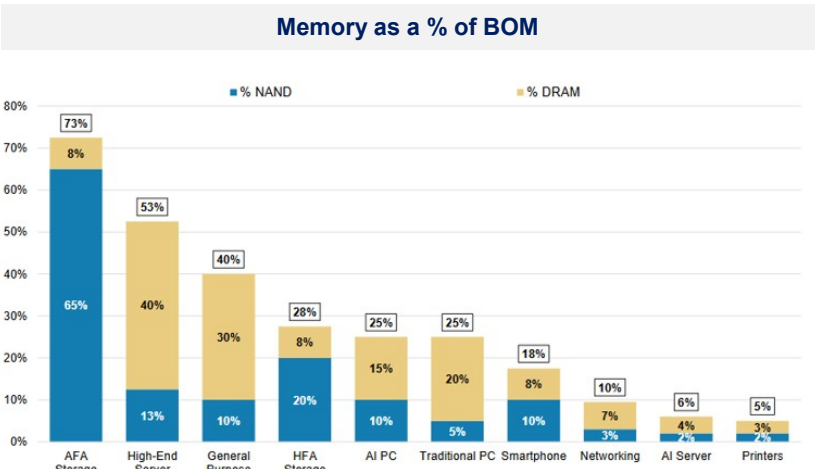
| Shipment (unit: k) | 1Q24   | 2Q24   | 3Q24   | 4Q24   | 1Q25   | 2Q25   | 3Q25   | 4Q25   | 1Q26   | 2Q26   | 2Q26E  | Diff. | 3Q26E  | 3Q26E  | Diff. |
|--------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-------|--------|--------|-------|
| Quanta             | 10,500 | 11,700 | 12,600 | 11,100 | 10,800 | 12,100 | 12,700 | 10,900 | 10,000 | 11,500 | 10,500 | 10%   | 11,000 | 10,800 | 2%    |
| Wistron            | 4,600  | 5,100  | 5,200  | 5,300  | 4,900  | 5,900  | 6,400  | 6,900  | 6,100  | 5,500  | 5,700  | -4%   | 4,700  | 5,500  | -15%  |
| Others             | 4,500  | 4,900  | 5,200  | 5,400  | 5,000  | 5,600  | 5,400  | 5,300  | 5,400  | 5,400  | 5,200  | 4%    | 5,000  | 4,500  | 11%   |
| Compal             | 7,500  | 8,700  | 8,300  | 7,800  | 7,000  | 7,100  | 7,100  | 6,800  | 5,900  | 6,200  | 6,000  | 3%    | 6,400  | 6,400  | 0%    |
| Pegatron           | 1,550  | 2,030  | 2,475  | 1,950  | 1,885  | 2,375  | 2,400  | 2,410  | 1,800  | 1,750  | 1,850  | -5%   | 1,800  | 1,800  | 0%    |
| Total              | 28,650 | 32,430 | 33,775 | 31,550 | 29,585 | 33,075 | 34,000 | 32,310 | 29,200 | 30,350 | 29,250 | 4%    | 28,900 | 29,000 | 0%    |
| <b>QoQ %</b>       |        |        |        |        |        |        |        |        |        |        |        |       |        |        |       |
| Quanta             | 1%     | 11%    | 8%     | -12%   | -3%    | 12%    | 5%     | -14%   | -8%    | 15%    | 5%     |       | -4%    | 3%     |       |
| Wistron            | -13%   | 11%    | 2%     | 2%     | -8%    | 20%    | 8%     | 8%     | -12%   | -10%   | -7%    |       | -15%   | -4%    |       |
| Others             | 0%     | 9%     | 6%     | 4%     | -7%    | 12%    | -4%    | -2%    | 2%     | 0%     | -4%    |       | -7%    | -13%   |       |
| Compal             | -10%   | 16%    | -5%    | -6%    | -10%   | 1%     | 0%     | -4%    | -13%   | 5%     | 2%     |       | 3%     | 7%     |       |
| Pegatron           | -6%    | 31%    | 22%    | -21%   | -3%    | 26%    | 1%     | 0%     | -25%   | -3%    | 3%     |       | 3%     | -3%    |       |
| Total              | -5%    | 13%    | 4%     | -7%    | -6%    | 12%    | 3%     | -5%    | -10%   | 4%     | 0%     |       | -5%    | -1%    |       |
| <b>YoY%</b>        |        |        |        |        |        |        |        |        |        |        |        |       |        |        |       |
| Quanta             | -3%    | -7%    | -4%    | 7%     | 3%     | 3%     | 1%     | -2%    | -7%    | -5%    | -13%   |       | -13%   | -15%   |       |
| Wistron            | 18%    | 11%    | 0%     | 0%     | 7%     | 16%    | 23%    | 30%    | 24%    | -7%    | -3%    |       | -27%   | -14%   |       |
| Others             | 2%     | 0%     | 6%     | 20%    | 11%    | 14%    | 4%     | -2%    | 8%     | -4%    | -7%    |       | -7%    | -17%   |       |
| Compal             | -1%    | 0%     | -11%   | -6%    | -7%    | -18%   | -14%   | -13%   | -16%   | -13%   | -15%   |       | -10%   | -10%   |       |
| Pegatron           | -9%    | -2%    | -1%    | 18%    | 22%    | 17%    | -3%    | 24%    | -5%    | -26%   | -22%   |       | -25%   | -25%   |       |
| Total              | 1%     | -1%    | -4%    | 5%     | 3%     | 2%     | 1%     | 2%     | -1%    | -8%    | -12%   |       | -15%   | -15%   |       |

Source: IDC, Morgan Stanley Research. e = Morgan Stanley Research estimates.

Memory Takes A Bite Out of Hardware Margins



Source: Morgan Stanley Research.

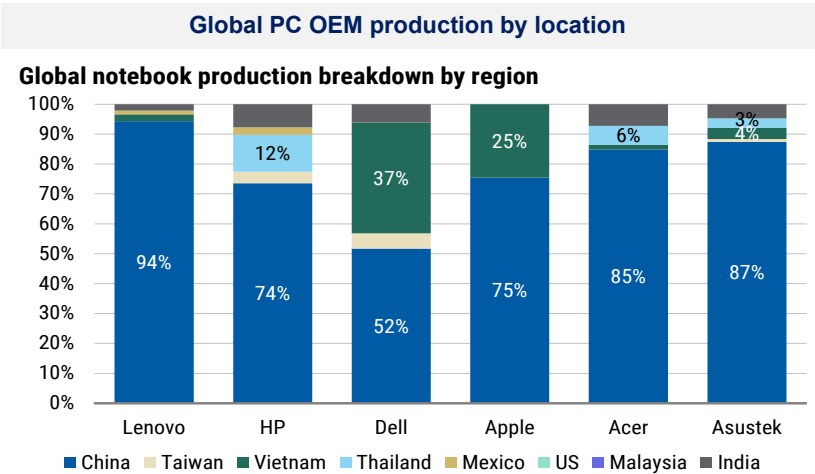
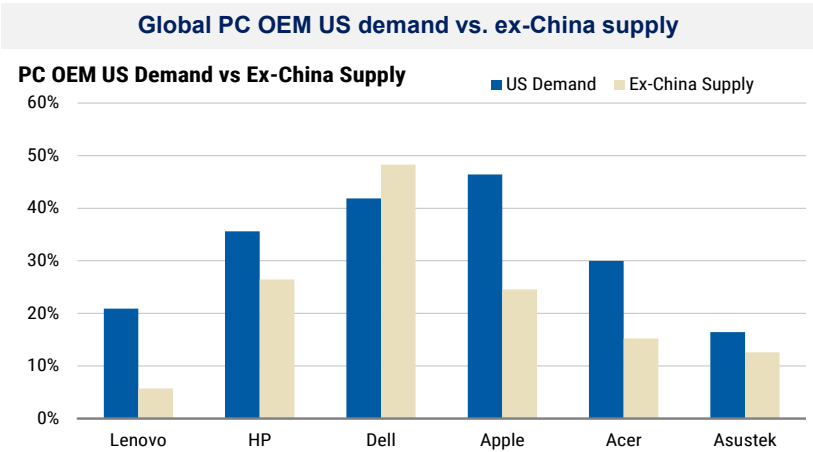
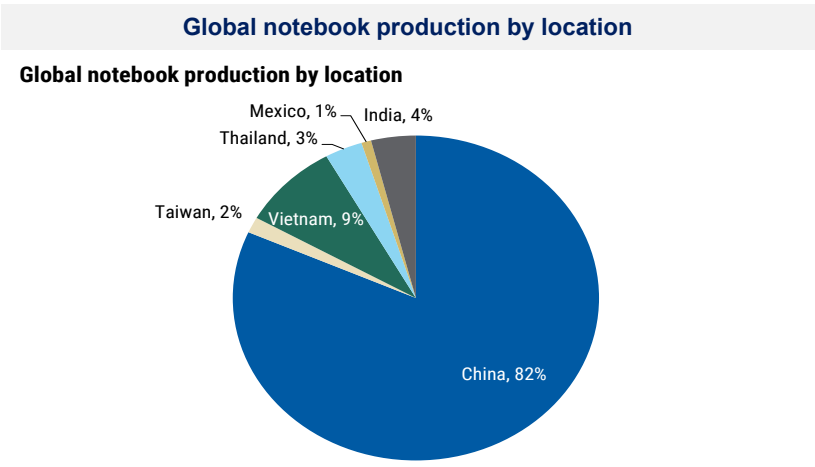
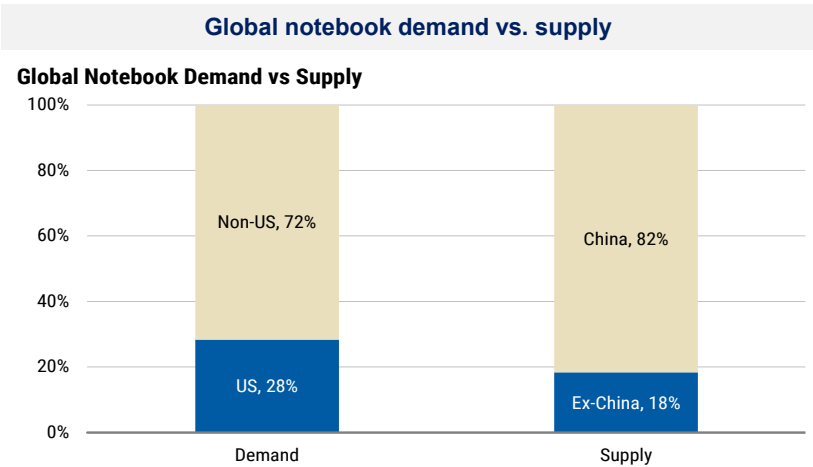


Asia IT Hardware OEM/ODM Scorecard

| Asia IT Hardware OEM/ODM Scorecard |                |         |                       |                   |                    |                 |
|------------------------------------|----------------|---------|-----------------------|-------------------|--------------------|-----------------|
|                                    | AI Beneficiary | Margins | Supply Chain Leverage | End-Market Demand | Special Situations | Memory Exposure |
| wiwynn                             | 4              | 4       | 4                     | 4                 | 1                  | 4               |
| wistron                            | 4              | 3       | 4                     | 3                 | 1                  | 4               |
| Quanta Computer                    | 4              | 3       | 4                     | 3                 | 1                  | 4               |
| Lenovo                             | 2              | 2       | 4                     | 2                 | 4                  | 1               |
| GIGABYTE                           | 3              | 3       | 3                     | 2                 | 1                  | 2.5             |
| ASUS                               | 2              | 3       | 2                     | 2                 | 1                  | 2.5             |
| PEGATRON                           | 2              | 1       | 2                     | 2                 | 1                  | 4               |
| COMPAL                             | 2              | 2       | 1                     | 2                 | 1                  | 4               |
| acer                               | 1              | 2       | 1                     | 1                 | 1                  | 2.5             |



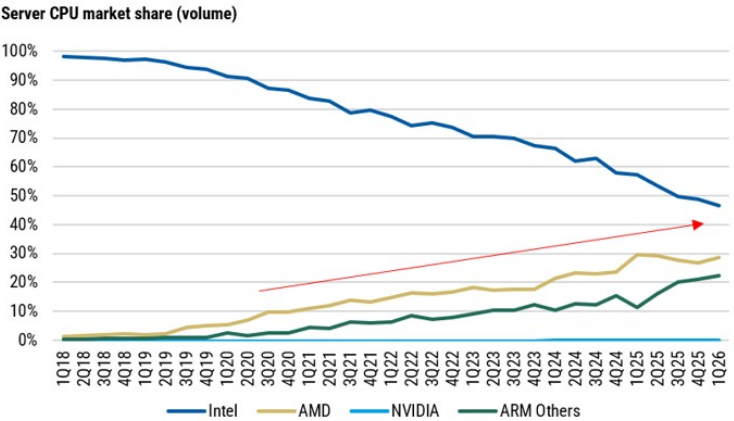
PC Market Overview



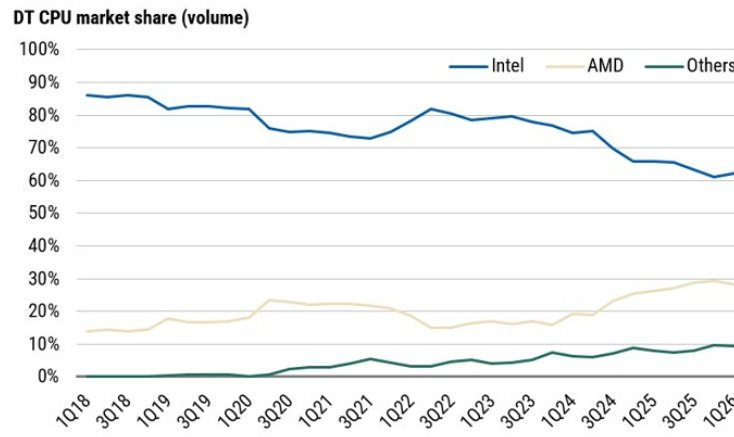
Source: IDC, Morgan Stanley Research. e = Morgan Stanley Research estimates. Note: Ex-China supply includes India production for India market.

CPU Market Share Trend

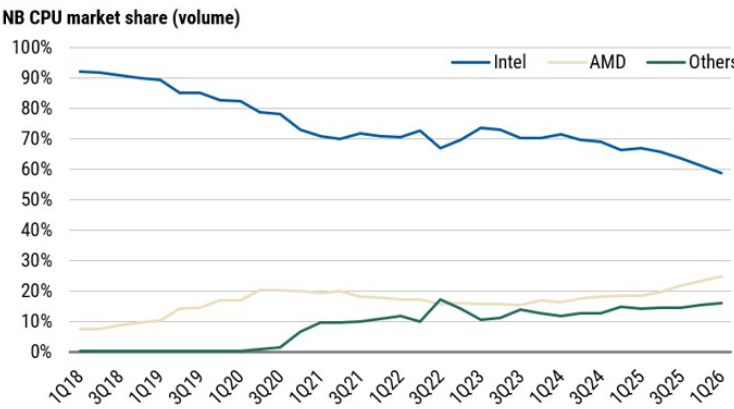
Server CPU market share



DT CPU market share



NB CPU market share



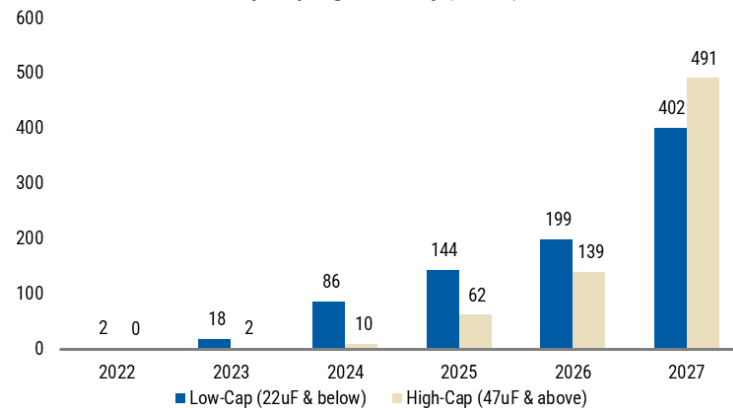
Source: IDC, Morgan Stanley Research.

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## Cloud and Edge AI Will Add ~US\$1B TAM to the Overall MLCC Market by 2027

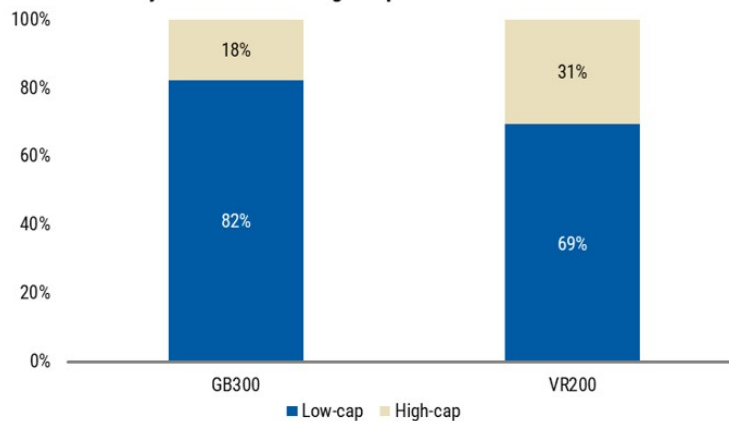
### Cloud AI to drive ~US\$1B TAM by 2026 (US\$ bn)

#### AI Server MLCC Demand Split by High/Low-Cap (US\$ M)



### AI Server will use significantly more high-cap (47uF+) MLCCs

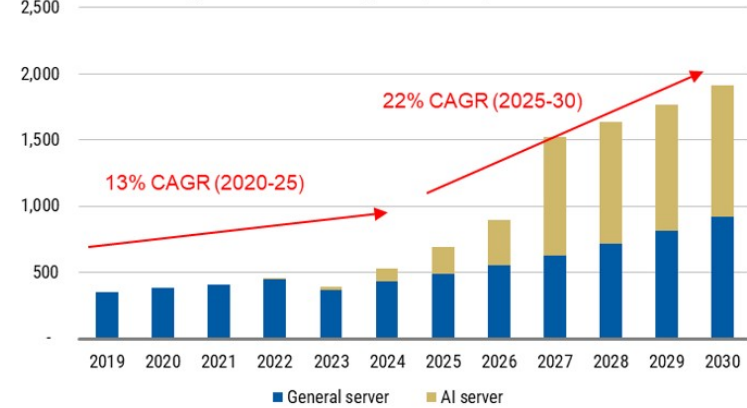
#### Nvidia's Rubin System Uses More High-Cap MLCCs



Source: IDC, Morgan Stanley Research. e = Morgan Stanley Research estimates.

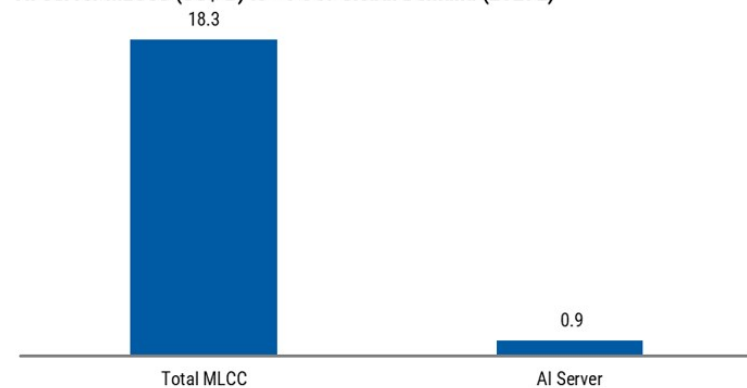
### AI Servers to drive significant MLCC TAM growth (US\$ M)

#### AI Servers to drive significant MLCC TAM growth (US\$ M)



### AI Server MLCC is ~3% of Global Units but ~5% of Global Values

#### AI Server MLCCs (US\$ B) is ~5% of Global Demand (2027E)



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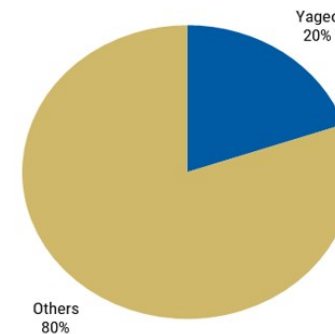
## AI Servers to See the Largest MLCC Content Growth

| Server Assumptions                   |      | # of MLCCs     |         | US\$         |       |
|--------------------------------------|------|----------------|---------|--------------|-------|
| <b>General server</b>                |      | <b>2,000</b>   |         | <b>30</b>    |       |
| <b>HGX Hopper AI server (8x GPU)</b> |      | <b>22,000</b>  |         | <b>190</b>   |       |
| GPU module (OAM)                     | 8x   | 1,300          | 10,400  | 10           | 80    |
| GPU baseboard (UBB)                  | 1x   | 4,900          | 4,900   | 45           | 45    |
| CPU motherboard                      | 1x   | 1,500          | 1,500   | 25           | 25    |
| Peripheral boards                    | 8x   | 650            | 5,200   | 5            | 40    |
| <b>HGX B200 AI server (8x GPU)</b>   |      | <b>36,000</b>  |         | <b>317</b>   |       |
| GPU module (OAM)                     | 8x   | 3,000          | 24,000  | 24           | 192   |
| GPU baseboard (UBB)                  | 1x   | 5,300          | 5,300   | 60           | 60    |
| CPU motherboard                      | 1x   | 1,500          | 1,500   | 25           | 25    |
| Peripheral boards                    | 8x   | 650            | 5,200   | 5            | 40    |
| <b>HGX B300 AI server (8x GPU)</b>   |      | <b>48,000</b>  |         | <b>435</b>   |       |
| GPU module (OAM)                     | 8x   | 4,400          | 35,200  | 40           | 320   |
| GPU baseboard (UBB)                  | 1x   | 6,100          | 6,100   | 50           | 50    |
| CPU motherboard                      | 1x   | 1,500          | 1,500   | 25           | 25    |
| Peripheral boards                    | 8x   | 650            | 5,200   | 5            | 40    |
| <b>GB200 (NVL72)</b>                 |      | <b>337,500</b> |         | <b>1,710</b> |       |
| Bianca                               | 36x  | 6,800          | 244,800 | 30           | 1,080 |
| Switch board                         | 9x   | 3,800          | 34,200  | 20           | 180   |
| Peripheral boards                    | 90x  | 650            | 58,500  | 5            | 450   |
| <b>GB300 (NVL72)</b>                 |      | <b>319,500</b> |         | <b>1,530</b> |       |
| Bianca                               | 36x  | 6,300          | 226,800 | 25           | 900   |
| Switch board                         | 9x   | 3,800          | 34,200  | 20           | 180   |
| Peripheral boards                    | 90x  | 650            | 58,500  | 5            | 450   |
| <b>VR200 (NVL72)</b>                 |      | <b>571,050</b> |         | <b>4,320</b> |       |
| Strata                               | 36x  | 11,000         | 396,000 | 90           | 3,240 |
| Switch board                         | 9x   | 7,000          | 63,000  | 45           | 405   |
| BlueField Module                     | 18x  | 2,000          | 36,000  | 5            | 90    |
| CX9 Orchid Module                    | 72x  | 650            | 46,800  | 5            | 360   |
| Peripheral boards                    | 45x  | 650            | 29,250  | 5            | 225   |
| <b>Vera Rack</b>                     |      | <b>494,048</b> |         | <b>3,060</b> |       |
| Vera MGX                             | 128x | 3,316          | 424,448 | 20           | 2,560 |
| Switch board                         | 4x   | 7,000          | 28,000  | 45           | 180   |
| Peripheral boards                    | 64x  | 650            | 41,600  | 5            | 320   |
| <b>STX Rack</b>                      |      | <b>326,256</b> |         | <b>2,415</b> |       |
| Mainboard                            | 16x  | 4,791          | 76,656  | 31           | 495   |
| Peripheral boards                    | 384x | 650            | 249,600 | 5            | 1,920 |
| <b>LPX Rack</b>                      |      | <b>581,024</b> |         | <b>4,326</b> |       |
| LPX UBB                              | 16x  | 5,374          | 85,984  | 35           | 559   |
| LPX Module                           | 256x | 1,705          | 436,480 | 14           | 3,527 |
| BF4                                  | 16x  | 2,000          | 32,000  | 5            | 80    |
| x86 CPU Module                       | 16x  | 1,660          | 26,560  | 10           | 160   |

Source: Company data. e = Morgan Stanley Research estimates.

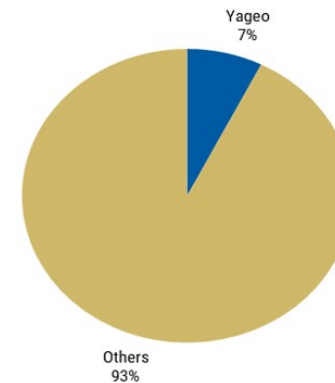
General server MLCC supplier share (value)

Yageo general server MLCC supplier share (value)



General PC MLCC supplier share (value)

Yageo general PC MLCC supplier share (value)

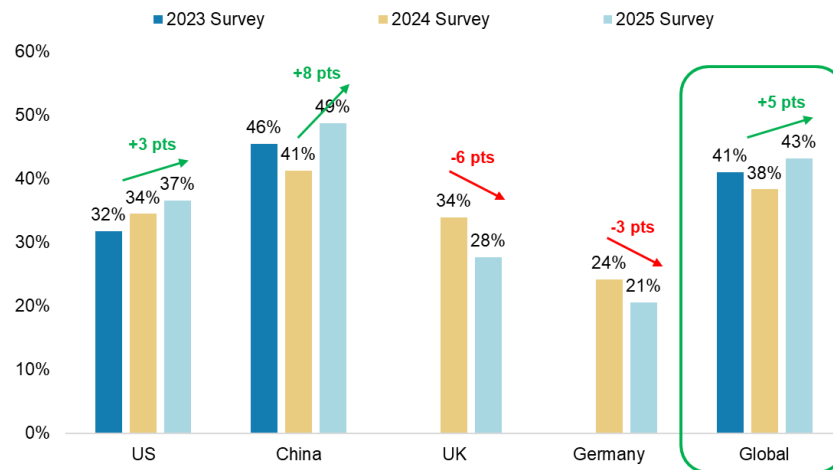


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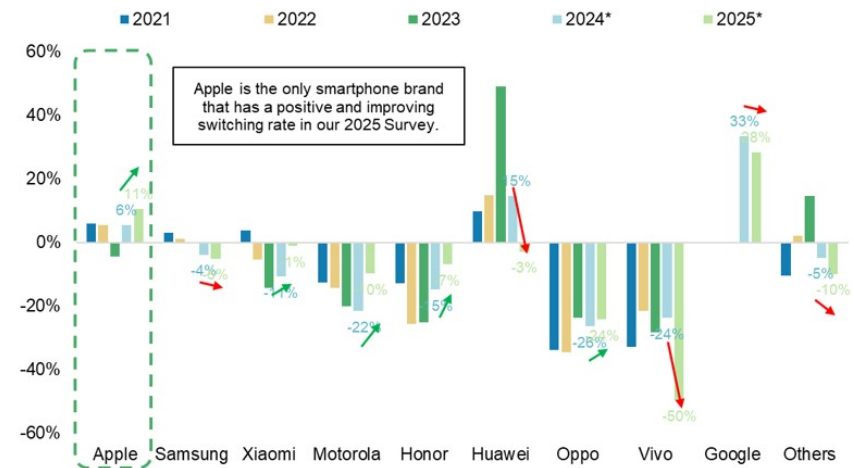
## All-time Survey-High Smartphone Upgrade Rates, Led Primarily By iPhone

- Based on our 2025 AlphaWise Smartphone Survey, next 12-month China and US smartphone upgrade rates are expected to expand 8 points and 3 points Y/Y, reaching an all-time survey (10+ years) high, while the same metrics are expected to decline 3-6 points Y/Y for EMEA (the UK and Germany).
- On a global blended basis, smartphone upgrade rates are expected to improve 5ppt Y/Y to 43%, an all-time survey high.
- However, Apple is the only smartphone OEM with positive and improving net switching rates in our 2025 AlphaWise Smartphone Survey, suggesting the upgrade rate momentum is primarily driven by iPhone strength.

Smartphone Upgrade Rates (Extremely Likely Upgraders) By Country



Global Smartphone Brand Net Switching Rates (The Higher The Better)



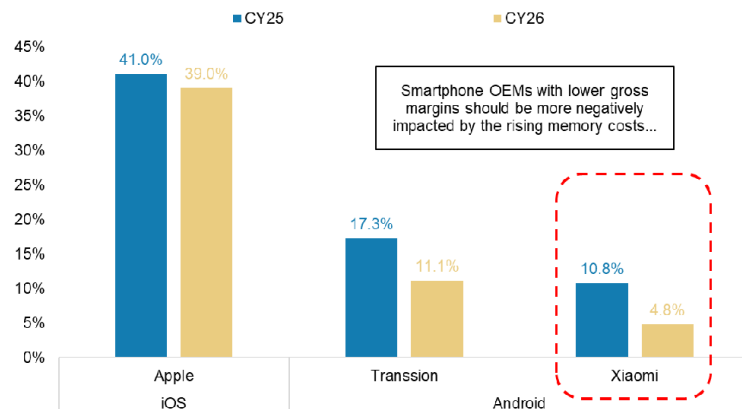
Source: AlphaWise, Morgan Stanley Research estimates. Note: 2024 and 2025 Survey results include UK and Germany (along with US and China) for the blended average; we believe Survey results in a certain year have implications to the estimates for the following year.

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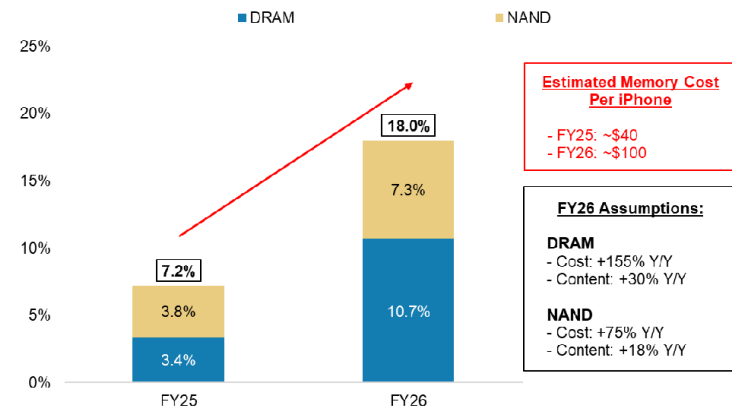
## The Reality Is – Smartphone OEMs Need To Pay for Higher Memory Prices

- Combining our checks and the Survey results on iPhone storage mix, we estimate average memory cost per iPhone will rise 150%+ Y/Y in 2026.
- This means that even for Apple, smartphone OEMs would need to pay significantly higher memory prices and raise prices to partially mitigate the cost impact.
- We believe this is particularly negative to Android players – smartphone gross margins are lower, the mix of memory cost within COGS is higher, and the customer base is more elastic (i.e., more sensitive to price increases).

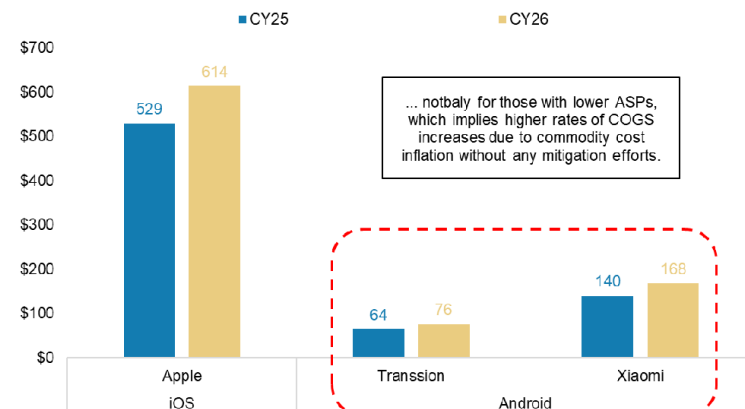
Smartphone Gross Margin By OEM



Memory Costs % of iPhone COGS



Smartphone COGS By OEM



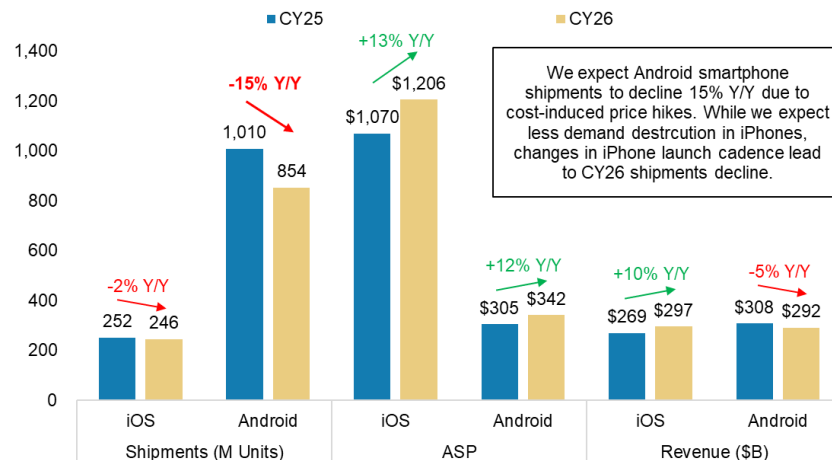
Source: AlphaWise, Morgan Stanley Research estimates.

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## Combined, Global Smartphone Units Are Expected to Decline 11% Y/Y, Driven By Android

- We believe price hikes by smartphone brands are inevitable. This will lead to demand destruction, notably in the mid-range Android smartphone market, which faces a more demand elastic customer base. We forecast Android smartphone shipments to decline 15% Y/Y in 2026.
- While the iPhone customer base is much less elastic than that of Android, we still expect shipments to decline in 2026, due primarily to changes in iPhone launch cadence.
- Taken together, we expect global smartphone shipments to decline 13% Y/Y in 2026.**

iOS vs. Android: Shipments, ASP, and Revenue In CY25-CY26



## SMARTPHONE MODEL

## Units (M)

Rest of Asia/Pacific  
PRC  
EMEA  
North America  
Latin America  
India  
Global Total

|                      | CY26 Est. |       |       | CY27 Est. |       |       |
|----------------------|-----------|-------|-------|-----------|-------|-------|
|                      | New       | Old   | Diff. | New       | Old   | Diff. |
| Rest of Asia/Pacific | 157       | 181   | -13%  | 162       | 183   | -12%  |
| PRC                  | 246       | 284   | -14%  | 252       | 287   | -12%  |
| EMEA                 | 311       | 363   | -14%  | 316       | 363   | -13%  |
| North America        | 135       | 149   | -9%   | 142       | 149   | -5%   |
| Latin America        | 115       | 139   | -18%  | 118       | 141   | -16%  |
| India                | 136       | 182   | -25%  | 147       | 189   | -22%  |
| Global Total         | 1,100     | 1,299 | -15%  | 1,137     | 1,312 | -13%  |

## SMARTPHONE MODEL

## Y/Y Unit Growth

Rest of Asia/Pacific  
PRC  
EMEA  
North America  
Latin America  
India  
Global Total

|                      | CY26 Est. |      |          | CY27 Est. |      |        |
|----------------------|-----------|------|----------|-----------|------|--------|
|                      | New       | Old  | Diff.    | New       | Old  | Diff.  |
| Rest of Asia/Pacific | -14%      | 0%   | -14.2pts | 3%        | 1%   | 1.9pts |
| PRC                  | -14%      | 1%   | -14.2pts | 3%        | 1%   | 1.6pts |
| EMEA                 | -15%      | 0%   | -15.1pts | 2%        | 0%   | 1.8pts |
| North America        | -6%       | 5%   | -10.9pts | 5%        | 0%   | 4.7pts |
| Latin America        | -13%      | 6%   | -19.5pts | 3%        | 1%   | 2.2pts |
| India                | -10%      | 14%  | -24.7pts | 8%        | 4%   | 3.8pts |
| Global Total         | -12.8%    | 3.1% | -16.0pts | 3.4%      | 1.0% | 2.4pts |

## SMARTPHONE MODEL

## Y/Y ASP Growth

## Revenue (M)

## Y/Y Revenue Growth

|                    | CY26 Est.  |            |         | CY27 Est.  |            |        |
|--------------------|------------|------------|---------|------------|------------|--------|
|                    | New        | Old        | Diff.   | New        | Old        | Diff.  |
| Y/Y ASP Growth     | \$535      | \$482      | 11%     | \$573      | \$501      | 14%    |
| Revenue (M)        | 17%        | 6%         | 11.0pts | 7%         | 4%         | 3.3pts |
| Y/Y Revenue Growth | \$ 588,225 | \$ 626,102 | -6%     | \$ 651,991 | \$ 657,405 | -1%    |
|                    | 2%         | 10%        | -8.0pts | 11%        | 5%         | 5.8pts |

Source: IDC, Morgan Stanley Research estimates. Note: iOS (iPhone) ASP and revenue here could be different from our AAPL model as the figures here are based on IDC's revenue estimates in CY25 along with our iPhone ASP growth forecasts in CY26.

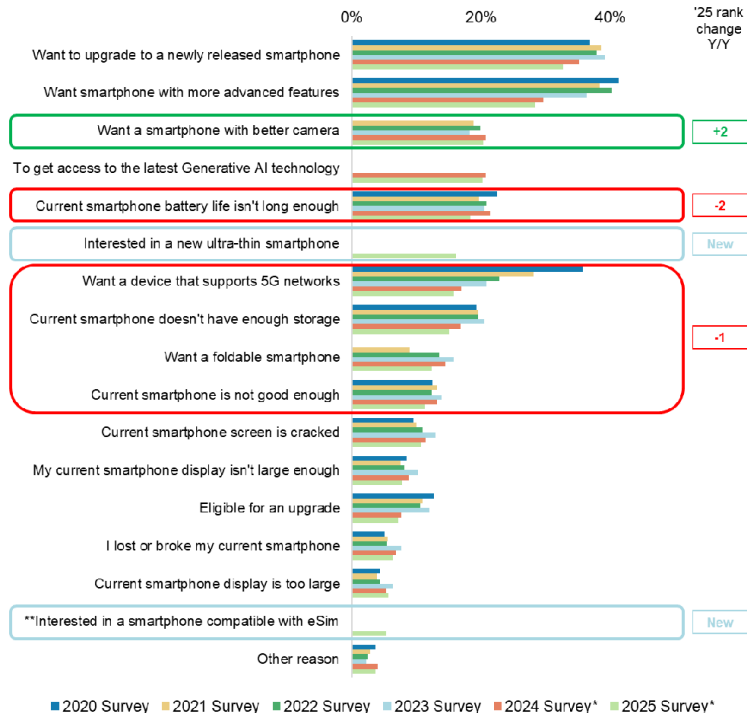


Morgan Stanley

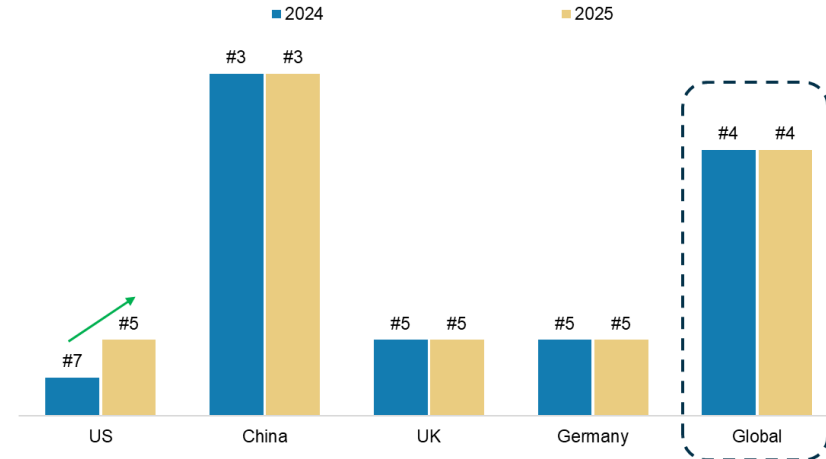
## Alongside Battery Life & Camera, Software Is Becoming a More Important Upgrade Factor

- Similar to last year, we continue to see a focus on new models, advanced features, and access to Gen AI technologies. What is slightly different is increasing interest in better cameras, alongside deteriorating interest in battery life and storage.
- AI remains a Top 5 reason for upgrading a smartphone, but without major advances in Edge AI or truly personalized digital assistants, we are not seeing much change in interest outside of China.

Global Smartphone: Reason for Upgrade (Among Smartphone Owners)



Access to AI Tech as The Ranking of Reasons to Upgrade Smartphone



Given rising memory costs, we would also expect smartphone OEMs to reduce BOM – based on our Survey results, battery and storage could be areas that are at risk of de-specing.

Source: AlphaWise, Morgan Stanley Research estimates. Note: 2024 and 2025 Survey results include UK and Germany (along with US and China) for the blended average; "Interested in a smartphone compatible with eSIM" was only asked in China.

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## We Favor Apple Supply Chain Over Android in 2026

- We expect the Android smartphone supply chain to be negatively affected by more meaningful volume declines, de-specing, and margin pressures, as OEMs attempt to save costs while raising prices.
- While we don't expect to see iPhone de-specing, we do expect Apple to share increasing costs with non-memory suppliers while keeping prices unchanged for the iPhone 17 family. With robust shipments and a favorable model mix, Apple and its suppliers are in a better position (vs. Android counterparts) within the smartphone universe.
- Consistent with our 2024 AlphaWise Survey takeaways, we maintain a relative preference for Apple/Apple supply chain over Android OEMs/supply chain.

| Most Favored |              |           | Least Favored |               |           |
|--------------|--------------|-----------|---------------|---------------|-----------|
| Ticker       | Company      | MS Rating | Ticker        | Company       | MS Rating |
| 005930.KS    | Samsung      | OW        | 3105.TW       | WIN Semi      | UW        |
| 2317.TW      | Hon Hai      | OW        | 300782.SZ     | Maxscend      | UW        |
| 0285.HK      | BYDE         | OW        | 8086.TWO      | AWSC          | UW        |
| 601138.SS    | FII          | OW        | 3034.TW       | Novatek       | UW        |
| 002475.SZ    | Luxshare     | OW        | 603160.SS     | Goodix        | UW        |
| 6770.T       | Alps Alpine  | OW        | 2382.HK       | Sunny Optical | EW        |
| 6762.T       | TDK          | OW        | QCOM          | Qualcomm      | UW        |
| 6088.HK      | FIT Hon Teng | OW        | QRVO          | Qorvo         | EW        |
| SWKS         | Skyworks     | EW        | 688036.SS     | Transsion     | EW        |
| AAPL         | Apple        | OW        | 603501.SS     | OmniVision    | EW        |
| 3017.TW      | AVC          | OW        |               |               |           |
| 6981.T       | Murata       | OW        |               |               |           |
| 2018.HK      | AAC          | OW        |               |               |           |
| 2454.TW      | MediaTek     | OW        |               |               |           |
| 601231.SS    | USI          | OW        |               |               |           |
| STMPA.PA     | STMicro      | OW        |               |               |           |

Source: FactSet, Morgan Stanley Research.

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## 3Q26 iPhone Build Estimates Kept at 54M Units (+4% q/q, -2% y/y)

| mn units                       | 1Q23        | 2Q23        | 3Q23        | 4Q23        | 2023         | 1Q24        | 2Q24        | 3Q24        | 4Q24        | 2024         | 1Q25        | 2Q25        | 3Q25        | 4Q25        | 2025         | 1Q26        | 2Q26e       | 3Q26e       |
|--------------------------------|-------------|-------------|-------------|-------------|--------------|-------------|-------------|-------------|-------------|--------------|-------------|-------------|-------------|-------------|--------------|-------------|-------------|-------------|
| <b>iPhone Sell-in</b>          |             |             |             |             |              |             |             |             |             |              |             |             |             |             |              |             |             |             |
| iPhone SE3 / SE 5G / 16e / 17e | 1.0         | 2.5         | 5.0         | 3.0         | 11.5         | 1.0         | 2.0         | 5.0         | 3.0         | 11.0         | 4.0         | 9.0         | 4.0         | 1.0         | 18.0         | 2.0         | 10.0        | 8.0         |
| iPhone XR                      | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone 11                      | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone 12 mini (5.4")          | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone 12 (6.1")               | 0.5         | 0.5         | -           | -           | 1.0          | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone 12 Pro (6.1")           | 0.5         | -           | -           | -           | 0.5          | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone 12 Pro Max (6.7")       | 0.5         | -           | -           | -           | 0.5          | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone 13 mini (5.4")          | 1.0         | 0.5         | -           | -           | 1.5          | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone 13 (6.1")               | 1.0         | 0.5         | -           | -           | 1.5          | -           | -           | -           | -           | -            | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone 13 Pro (6.1")           | 1.0         | 2.0         | 1.0         | -           | 4.0          | 0.5         | 0.5         | -           | -           | 1.0          | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone 13 Pro Max (6.7")       | 2.0         | 2.0         | 1.0         | -           | 5.0          | 0.5         | 0.5         | -           | -           | 1.0          | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone 14 (6.1")               | 7.0         | 4.0         | 5.0         | 2.0         | 18.0         | 1.0         | 0.5         | -           | -           | 1.5          | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone 14 Plus (6.7")          | 5.0         | 3.0         | 1.0         | 1.0         | 10.0         | 1.0         | 0.5         | -           | -           | 1.5          | -           | -           | -           | -           | -            | -           | -           | -           |
| iPhone 14 Pro (6.1")           | 15.0        | 12.0        | 7.0         | 3.0         | 37.0         | 2.0         | 2.0         | 1.0         | 1.0         | 6.0          | 1.0         | 1.0         | 1.0         | -           | 3.0          | -           | -           | -           |
| iPhone 14 Pro Max (6.7")       | 19.5        | 14.0        | 9.0         | 4.0         | 46.5         | 2.0         | 2.0         | 1.0         | 1.0         | 6.0          | 1.0         | 1.0         | 1.0         | -           | 3.0          | -           | -           | -           |
| iPhone 15 (6.1")               |             |             | 6.0         | 12.0        | 18.0         | 6.0         | 5.0         | 4.0         | 1.0         | 16.0         | 1.0         | 0.5         | 0.5         | 0.5         | 2.5          | -           | -           | -           |
| iPhone 15 Plus (6.7")          |             |             | 5.0         | 10.0        | 15.0         | 4.0         | 3.0         | 1.0         | 1.0         | 9.0          | 1.0         | 0.5         | 0.5         | 0.5         | 2.5          | -           | -           | -           |
| iPhone 15 Pro (6.1")           |             |             | 8.0         | 16.0        | 24.0         | 12.0        | 10.0        | 8.5         | 2.0         | 32.5         | 2.0         | 2.0         | 2.0         | 1.0         | 7.0          | 1.0         | 1.0         | -           |
| iPhone 15 Pro Max (6.7")       |             |             | 2.0         | 24.0        | 26.0         | 18.0        | 13.0        | 10.5        | 3.0         | 44.5         | 3.0         | 3.0         | 2.0         | 1.0         | 9.0          | 1.0         | 1.0         | -           |
| iPhone 16 (6.1")               |             |             |             |             |              | -           | -           | 6.0         | 12.0        | 18.0         | 5.0         | 4.0         | 4.0         | 1.0         | 14.0         | 2.0         | 2.0         | 1.0         |
| iPhone 16 Plus (6.7")          |             |             |             |             |              | -           | -           | 5.0         | 6.0         | 11.0         | 4.0         | 3.0         | 1.0         | -           | 8.0          | -           | -           | -           |
| iPhone 16 Pro (6.3")           |             |             |             |             |              | -           | -           | 8.0         | 17.0        | 25.0         | 11.5        | 9.5         | 4.0         | 2.0         | 27.0         | 2.0         | 2.0         | 2.0         |
| iPhone 16 Pro Max (6.9")       |             |             |             |             |              | -           | -           | 4.0         | 26.0        | 30.0         | 16.5        | 13.0        | 10.0        | 4.0         | 43.5         | 5.0         | 3.0         | 2.0         |
| iPhone 17 (6.1")               |             |             |             |             |              | -           | -           | -           | -           | -            | -           | -           | 6.0         | 16.0        | 22.0         | 10.0        | 8.0         | 6.0         |
| iPhone Air (6.7")              |             |             |             |             |              | -           | -           | -           | -           | -            | -           | -           | 4.0         | 4.0         | 8.0          | -           | -           | -           |
| iPhone 17 Pro (6.3")           |             |             |             |             |              | -           | -           | -           | -           | -            | -           | -           | 8.0         | 22.0        | 30.0         | 18.0        | 13.0        | 9.0         |
| iPhone 17 Pro Max (6.9")       |             |             |             |             |              | -           | -           | -           | -           | -            | -           | -           | 7.0         | 23.0        | 30.0         | 15.0        | 12.0        | 9.0         |
| iPhone 18 Pro (6.3")           |             |             |             |             |              |             |             |             |             |              |             |             |             |             |              |             |             | 8.0         |
| iPhone 18 Pro Max (6.9")       |             |             |             |             |              |             |             |             |             |              |             |             |             |             |              |             |             | 8.0         |
| iPhone Fold                    |             |             |             |             |              |             |             |             |             |              |             |             |             |             |              |             |             | 1.0         |
| <b>Total</b>                   | <b>54.0</b> | <b>41.0</b> | <b>50.0</b> | <b>75.0</b> | <b>220.0</b> | <b>48.0</b> | <b>39.0</b> | <b>54.0</b> | <b>73.0</b> | <b>214.0</b> | <b>50.0</b> | <b>46.5</b> | <b>55.0</b> | <b>76.0</b> | <b>227.5</b> | <b>56.0</b> | <b>52.0</b> | <b>54.0</b> |
| YoY                            | 0%          | -9%         | 0%          | -3%         | -3%          | -11%        | -5%         | 8%          | -3%         | -3%          | 4%          | 19%         | 2%          | 4%          | 6%           | 12%         | 12%         | -2%         |
| QoQ                            | -30%        | -24%        | 22%         | 50%         |              | -36%        | -19%        | 38%         | 35%         |              | -32%        | -7%         | 18%         | 38%         |              | -26%        | -7%         | 4%          |

- For full-year 2026, our supply-chain checks suggest total iPhone build volume could be up in the low to mid single digits YoY.
- For foldable iPhone model in 2H26, our checks suggest sampling production of mechanical components already kicked off in March and still target launch in September.

Source: Company data, Morgan Stanley Research. e = Morgan Stanley Research estimates.

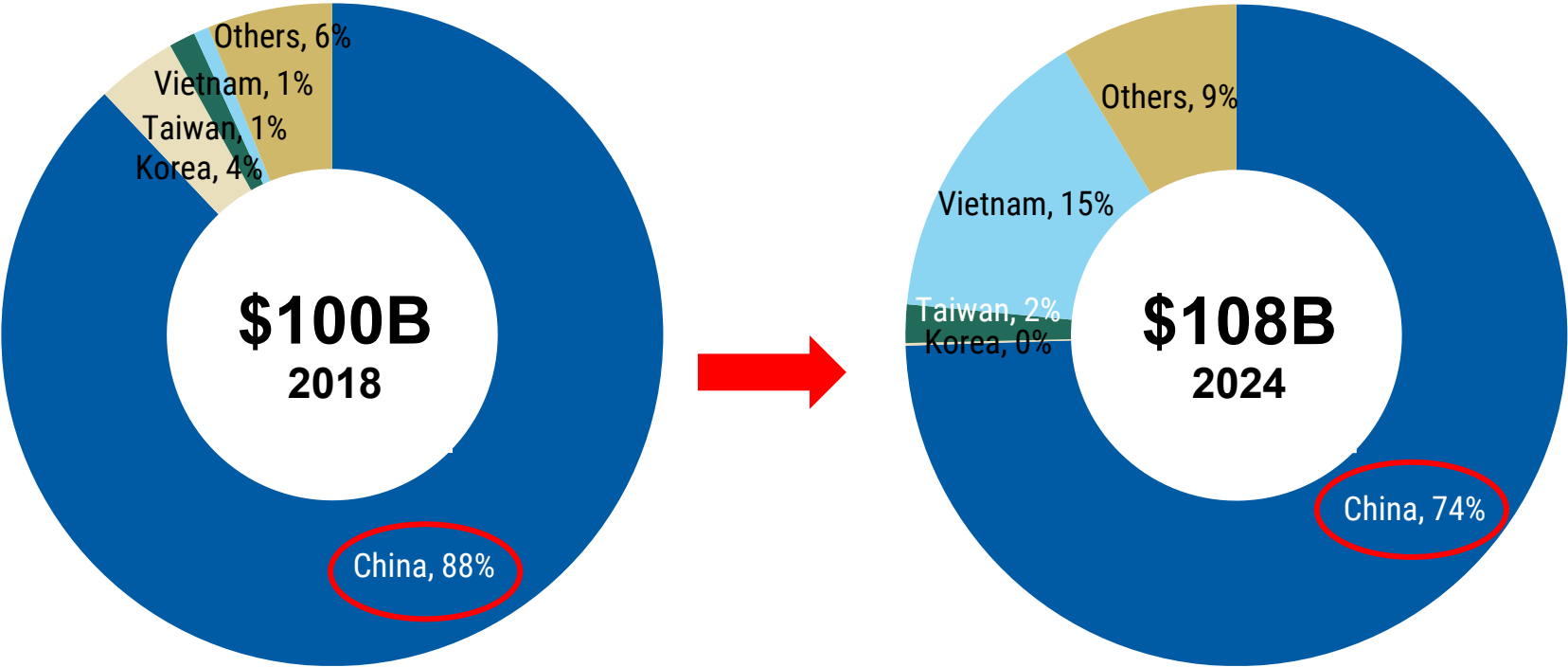
# Supply Chain Reorientation



## USMCA

Servers and switches produced in Mexico are considered **USMCA**-compliant goods and therefore exempt from tariffs. The current USMCA expires in **July 2036**.

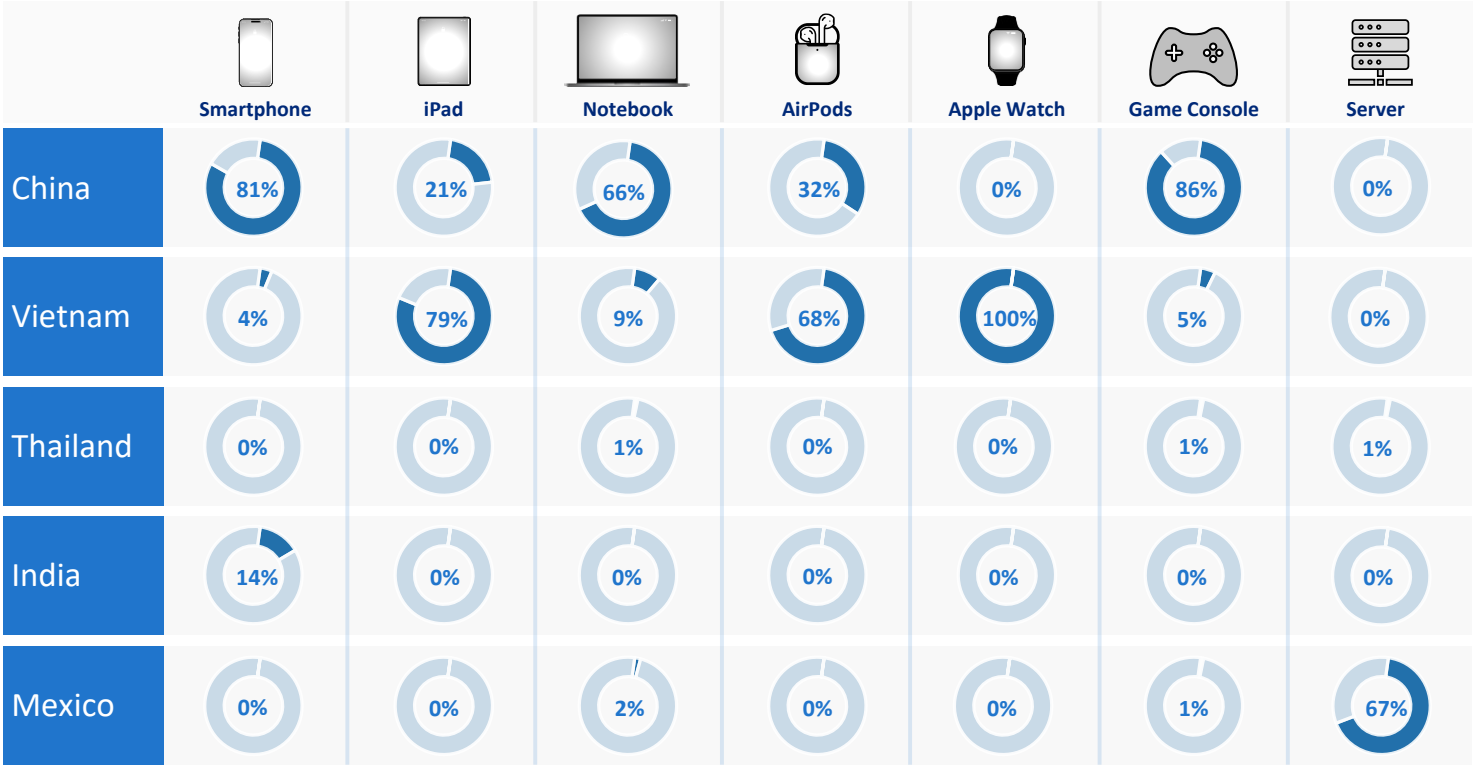
US Imports Data Shows Supply Chain Reorientation Is Happening



**Supply chain reorientation is happening:** In 2018, China accounted for 88% of US tech imports (notebooks, smartphones and game consoles) or ~US\$88bn, but by 2024, China dropped to 74%, or ~US\$80bn.

Source: US Comtrade, Morgan Stanley Research.  
Note: Import data includes notebook, smartphones, and game consoles.

# How Much USA Imports from Different Countries by Major Tech Products



US still imports the majority of smartphones, notebooks, and game consoles from China, but Vietnam has become the main exporter of iPads, AirPods, and Apple Watches to the US. We are seeing more notebook production ramping up in Vietnam and Thailand, so these two countries will become more important for the US in the coming years. And for smartphones, we expect India to become a bigger contributor in the coming years as well. And for servers, we think Mexico will continue to be important as servers imported from Mexico into the US are USMCA-eligible for 0% tariff.

Source: US Comtrade, Morgan Stanley Research.

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(as of June 30, 2026)

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| Stock Rating Category | Coverage Universe |            | Investment Banking Clients (IBC) |                |                      | Other Material Investment Services Clients (MISC) |                       |
|-----------------------|-------------------|------------|----------------------------------|----------------|----------------------|---------------------------------------------------|-----------------------|
|                       | Count             | % of Total | Count                            | % of Total IBC | % of Rating Category | Count                                             | % of Total Other MISC |
| Overweight/Buy        | 1544              | 42%        | 453                              | 49%            | 29%                  | 757                                               | 44%                   |
| Equal-weight/Hold     | 1577              | 43%        | 390                              | 42%            | 25%                  | 769                                               | 44%                   |
| Not-Rated/Hold        | 3                 | 0%         | 1                                | 0%             | 33%                  | 1                                                 | 0%                    |
| Underweight/Sell      | 544               | 15%        | 89                               | 10%            | 16%                  | 204                                               | 12%                   |
| Total                 | 3,668             |            | 933                              |                |                      | 1731                                              |                       |

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.



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## INDUSTRY COVERAGE: Greater China Technology Hardware

| COMPANY (TICKER)                                     | RATING (AS OF) | PRICE* (07/14/2026) |
|------------------------------------------------------|----------------|---------------------|
| <b>Andy Meng, CFA</b>                                |                |                     |
| AAC Technologies Holdings (2018.HK)                  | O (01/29/2024) | HK\$38.00           |
| Accelink Technologies Co. Ltd. (002281.SZ)           | U (05/12/2022) | Rmb232.95           |
| BYD Electronics (0285.HK)                            | O (04/28/2023) | HK\$23.24           |
| China TransInfo Technology Co Ltd (002373.SZ)        | E (07/18/2023) | Rmb7.03             |
| Dahua Technology Co. Ltd. (002236.SZ)                | E (12/12/2024) | Rmb16.32            |
| Eoptolink Technology Inc Ltd (300502.SZ)             | O (02/27/2026) | Rmb568.82           |
| Genius Electronic Optical Co. Ltd. (3406.TW)         | E (04/23/2025) | NT\$587.00          |
| GoerTek Inc (002241.SZ)                              | U (04/23/2025) | Rmb21.73            |
| Gosuncn Technology Group Co Ltd (300098.SZ)          | U (11/07/2022) | Rmb4.30             |
| HIKVision Digital Technology (002415.SZ)             | E (12/12/2024) | Rmb33.58            |
| Largan Precision (3008.TW)                           | O (07/10/2026) | NT\$4,265.00        |
| LianChuang Electronic Technology Co Ltd (002036.SZ)  | U (06/12/2024) | Rmb6.10             |
| Lingyi Itech Guangdong Co (002600.SZ)                | U (04/23/2025) | Rmb12.97            |
| Luxshare Precision Industry Co., Ltd. (002475.SZ)    | O (10/24/2016) | Rmb61.67            |
| OFILM Group Co Ltd (002456.SZ)                       | U (06/12/2024) | Rmb8.91             |
| Q Technology (Group) Company Ltd (1478.HK)           | U (03/23/2026) | HK\$6.77            |
| Quectel Wireless Solutions Co Ltd (603236.SS)        | E (10/09/2024) | Rmb50.45            |
| Shanghai Conant Optical Co Ltd (2276.HK)             | O (04/10/2026) | HK\$30.48           |
| Shenzhen Transsion Holdings Co Ltd (688036.SS)       | E (03/23/2026) | Rmb62.40            |
| Sunny Optical (2382.HK)                              | E (03/23/2026) | HK\$55.80           |
| Suzhou TFC Optical Communication Co Ltd. (300394.SZ) | E (11/06/2025) | Rmb273.51           |
| Wingtech Technology Co Ltd (600745.SS)               | U (05/18/2026) | Rmb19.95            |
| Xiaomi Corp (1810.HK)                                | O (04/14/2021) | HK\$26.04           |
| Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)  | U (10/13/2021) | Rmb400.18           |
| Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)    | O (07/15/2026) | HK\$153.90          |
| Yongxin Optics Co Ltd (603297.SS)                    | E (11/15/2022) | Rmb110.50           |

|                                                  |                |              |
|--------------------------------------------------|----------------|--------------|
| Zhejiang Crystal-Optech Co Ltd (002273.SZ)       | O (11/15/2022) | Rmb31.26     |
| Zhongji Innolight Co Ltd (300308.SZ)             | ++             | Rmb1,184.05  |
| ZTE Corporation (0763.HK)                        | O (06/04/2026) | HK\$24.78    |
| ZTE Corporation (000063.SZ)                      | E (06/04/2026) | Rmb39.91     |
| <b>Derrick Yang</b>                              |                |              |
| Accton Technology Corporation (2345.TW)          | O (06/06/2024) | NT\$2,280.00 |
| Advantech (2395.TW)                              | O (01/20/2021) | NT\$529.00   |
| AirTAC International (1590.TW)                   | O (04/16/2025) | NT\$1,290.00 |
| Asia Vital Components Co. Ltd. (3017.TW)         | O (07/30/2024) | NT\$2,120.00 |
| AU Optronics (2409.TW)                           | E (02/10/2026) | NT\$27.55    |
| Auras Technology Co Ltd (3324.TWO)               | E (05/04/2023) | NT\$864.00   |
| Bizlink (3665.TW)                                | O (03/10/2025) | NT\$1,775.00 |
| BOE Technology (000725.SZ)                       | O (09/06/2019) | Rmb7.02      |
| Chenbro (8210.TW)                                | O (07/23/2025) | NT\$1,110.00 |
| Chroma Ate Inc. (2360.TW)                        | O (10/05/2021) | NT\$1,820.00 |
| Delta Electronics Inc. (2308.TW)                 | O (07/13/2017) | NT\$1,855.00 |
| E Ink Holdings Inc. (8069.TWO)                   | O (05/11/2026) | NT\$193.50   |
| Ennostar Inc (3714.TW)                           | U (09/23/2022) | NT\$59.30    |
| Fositek Corp (6805.TW)                           | O (06/25/2025) | NT\$1,320.00 |
| Hiwin Technologies Corp. (2049.TW)               | O (03/30/2026) | NT\$317.00   |
| Innolux (3481.TW)                                | E (04/07/2025) | NT\$57.20    |
| King Slide Works Co. Ltd. (2059.TW)              | O (11/08/2023) | NT\$7,995.00 |
| Lens Technology (300433.SZ)                      | E (07/22/2020) | Rmb40.32     |
| Lite-On Technology (2301.TW)                     | E (01/15/2025) | NT\$212.50   |
| Radiant Opto-Electronics Corporation (6176.TW)   | E (03/01/2024) | NT\$87.80    |
| TCL Corp. (000100.SZ)                            | E (04/07/2025) | Rmb5.13      |
| Tianma Microelectronics (000050.SZ)              | U (01/24/2018) | Rmb7.09      |
| Wuhan Jingce Electronic Group Co Ltd (300567.SZ) | E (11/26/2021) | Rmb297.10    |
| <b>Howard Kao</b>                                |                |              |
| Acer Inc. (2353.TW)                              | U (04/23/2025) | NT\$30.75    |
| Asustek Computer Inc. (2357.TW)                  | U (11/16/2025) | NT\$703.00   |
| Catcher Technology (2474.TW)                     | U (11/17/2025) | NT\$189.50   |
| Compal Electronics (2324.TW)                     | U (04/23/2025) | NT\$35.55    |
| FIT Hon Teng Ltd (6088.HK)                       | O (11/03/2025) | HK\$6.08     |
| Foxconn Industrial Internet Co. Ltd. (601138.SS) | O (07/10/2019) | Rmb65.60     |
| Giga-Byte Technology Co. Ltd. (2376.TW)          | E (11/16/2025) | NT\$334.00   |
| Gold Circuit Electronics Ltd. (2368.TW)          | O (10/06/2022) | NT\$1,015.00 |
| Hon Hai Precision (2317.TW)                      | O (03/15/2024) | NT\$235.50   |
| Inspur Electronic Information (000977.SZ)        | E (08/28/2023) | Rmb84.95     |
| LandMark Optoelectronics Corporation (3081.TWO)  | E (03/26/2026) | NT\$1,925.00 |

|                                           |                |              |
|-------------------------------------------|----------------|--------------|
| Lenovo (0992.HK)                          | O (07/10/2026) | HK\$23.16    |
| Lotes Co. Ltd. (3533.TW)                  | E (05/12/2025) | NT\$1,925.00 |
| Nan Ya PCB (8046.TW)                      | O (02/23/2026) | NT\$1,290.00 |
| Pegatron Corporation (4938.TW)            | E (03/25/2026) | NT\$82.10    |
| Quanta Computer Inc. (2382.TW)            | O (05/01/2023) | NT\$380.00   |
| Shengyi Technology Co Ltd. (600183.SS)    | E (05/26/2022) | Rmb147.90    |
| Shennan Circuits Co Ltd (002916.SZ)       | E (08/24/2023) | Rmb408.99    |
| Unimicron (3037.TW)                       | O (02/23/2026) | NT\$874.00   |
| Visual Photonics Epitaxy Co Ltd (2455.TW) | E (09/11/2023) | NT\$314.50   |
| Wistron Corporation (3231.TW)             | O (07/12/2023) | NT\$142.50   |
| Wiwynn Corp (6669.TW)                     | O (11/10/2025) | NT\$4,905.00 |
| Yageo Corp. (2327.TW)                     | O (10/28/2025) | NT\$778.00   |
| Zhen Ding (4958.TW)                       | O (05/18/2026) | NT\$629.00   |

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\* Historical prices are not split adjusted.